THE AREA FUNCTION OF THE MERTENS EXCURSION SKELETON AND HORIZON-LOCALIZED CRITERIA FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. We decompose the Mertens function $M(x) = \sum_{n \le x} \mu(n)$ into its excursions between zero runs and study a regularized triangular skeleton of these excursions, its area curves, and exact quadratic and correlation identities inside each excursion. Two elementary counting lemmas show that, among excursions starting at most X, those with $x_1 > x_0^2$ number $O(\log\log X)$, and those with $x_1 > \lambda x_0$ number $O(\log X/\log\lambda)$. Combined with the Kaczorowski–Pintz lower bound for large alternating oscillations, this yields a general localization theorem: the Riemann hypothesis is equivalent to Littlewood's bound $M(t) \ll_\varepsilon t^{1/2+\varepsilon}$ restricted to the points t of each excursion lying below a horizon $h(x_0)$, whenever the excursions crossing the horizon are fewer than $(\gamma_1/\pi)\log X$. The horizon can be eroded from x_0^2 through the fractional square chain $x_0^2/2^j$, then through $x_0\psi(x_0)$ with $\psi \to \infty$, and finally down to λx_0 for every $\lambda > e^{\pi/\gamma_1}$. After deleting zero increments, a completed excursion is a primitive Dyck path on its fixed support mask. A pointwise support-time tail bound shows that, under the uniform primitive Catalan reference law on that mask, a value $|M(t)| \ge t^{1/2+\varepsilon}$ in an excursion with $x_1 \le \lambda x_0$ has probability at most $x_0^{O(1)} e^{-x_0^{2\varepsilon}/2}$. If RH fails, a sequence with $|M(t_j)|/t_j^{1/2+\varepsilon_0} \to \infty$ occurs in such excursions, with Catalan cost growing like a power of x_0. Polynomial lower bounds for these tail weights, for complete excursions or for open prefixes from the last zero, imply $|M(t)| \ll \sqrt{t\log t}$ on the corresponding ranges. On this localized linear window the relevant increment sum depends only on the Möbius prefix up to $\lambda x_0/2$. Thus RH admits a prefix-dependency cutoff $\theta_* = (1/2)e^{\pi/\gamma_1} = 0.6244\ldots$ in the sense that every $\theta > \theta_*$ suffices. We also isolate the exact intra-excursion constraints, give a ballistic counterexample to the corresponding relaxed divisor-sum cone, and introduce a multiplicative-layer filtration whose k-th prefix layer determines the $(k+1)$-st layer in the future window. A bounded number of layers fixes only a density-zero set of window increments, whose sum is nevertheless of order $u(\log\log u)^k/\log u$, so the RH scale requires cancellation across $\asymp \log\log u$ layers. The one-step recursion further yields a finite-channel scale cascade: above $\lambda u/(K+1)$ all dependence is concentrated in the $K-1$ scaled increments $M(t/k) - M(u/k)$, $2 \le k \le K$. A deterministic persistence lemma shows that a high intra-excursion value forces one of these channels, or the deeper remainder, to remain macroscopically positive at many points of the final ascent. Independently, the same counting bounds convert any uniform sub-Gaussian tail for a normalized excursion statistic into extreme scales $\sqrt{\log\log X}$ and $\sqrt{\log\log\log X}$ for the linear and square families respectively. We also give an exact bilinear representation of M on the square range of an excursion, an exact Mellin decomposition of $1/\zeta$ into excursion blocks, and an asymptotic analysis of the split of M into smooth and large-prime parts.

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1. Introduction

The object of this paper is the Mertens function M, decomposed into excursions between zero runs (Section 2). Two kinds of information about M are separated throughout: what is forced by the ordering and local geometry of the excursions alone, and what requires arithmetic input.

Main results. The central results are in Section 8.

Geometric counts. For any ordered family of disjoint integer intervals $(x_0, x_1]$, the members with $x_1 > x_0^A$ $(A > 1)$ starting below X are $O_A(\log\log X)$, those with $x_1 > cx_0^2$ are $O_c(\log\log X)$, those with $x_1 > x_0\psi(x_0)$ are $o(\log X)$ whenever $\psi \to \infty$, and those with $x_1 > \lambda x_0$ are at most $\log X/\log\lambda + O(1)$ (Lemmas 8.1, 8.2, 8.3, 8.4). No arithmetic enters.

Horizon localization. By Kaczorowski–Pintz [9], if $\Theta > 1/2$ then at least $(\gamma_1/\pi - o(1))\log X$ distinct excursions starting below X contain values of size $t^{\Theta-\eta}$. Theorem 8.11 turns every geometric count below this threshold into an equivalent form of RH: for any horizon h whose crossing family has at most $(\kappa + o(1))\log X$ members with $\kappa < \gamma_1/\pi$,

$$\mathrm{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \quad \text{for } x_0 < t \le \min\{x_1, h(x_0)\}.$$

In particular (Corollary 8.13)

$$\mathrm{RH} \iff \text{Littlewood's bound holds on } x_0 < t \le \min\{x_1, \lambda x_0\}, \qquad \lambda > \lambda_* := e^{\pi/\gamma_1} = 1.2488\ldots,$$

and the square horizon x_0^2 is eroded from above through cx_0^2 and $x_0\psi(x_0)$. If RH fails, a positive proportion (on the scale $\log X$) of the Kaczorowski–Pintz excursions carry their large value in $[x_0 + \frac{1}{2}x_0^{\Theta-\eta}, \lambda x_0]$ (Corollary 8.28). The threshold λ_* is the multiplicative distance between consecutive sign changes of $\cos(\gamma_1 \log x)$ (Remark 8.29). The fractional-square lemma makes the intermediate erosion explicit:

$$x_0^2 \longrightarrow \frac{x_0^2}{2} \longrightarrow \frac{x_0^2}{4} \longrightarrow \frac{x_0^2}{8} \longrightarrow \cdots \longrightarrow x_0\psi(x_0) \longrightarrow \lambda x_0.$$

For each fixed dyadic level the crossing family is $O(\log\log X)$; the passage to a level depending on x_0 is supplied by the superlinear-horizon lemma, not by a uniform use of the fixed-c estimate.

Kaczorowski–Pintz [9] provide the oscillation-density input. The excursion-counting lemmas, the horizon-localization theorem, the descending-horizon principle, the threshold $\lambda_* = e^{\pi/\gamma_1}$, and the resulting prefix-dependency cutoff $\theta_* = \frac{1}{2}e^{\pi/\gamma_1}$ are developed here from that input.

Linear window. For $u < t \le 2u$ one has $\mu(n) = -\sum_{d|n,\, d\le u} \mu(d)$. After horizon localization, if $u < t \le \lambda u < 2u$, the terms with $d > \lambda u/2$ have no multiple in $(u, t]$, and therefore

$$M(t) = -\sum_{d\le \lambda u/2} \mu(d)\Big(\Big\lfloor \frac{t}{d}\Big\rfloor - \Big\lfloor \frac{u}{d}\Big\rfloor\Big).$$

Thus every $\theta > \theta_* := \frac{1}{2}e^{\pi/\gamma_1}$ is an admissible prefix cutoff for the RH-equivalent excursion window; the convenient choice $\lambda = 5/4$ uses only $d \le 5u/8$. This is a descent of dependency, not of size. More sharply, for every fixed $K \ge 2$ the dependence above $\lambda u/(K+1)$ is concentrated in the $K-1$ increments $M(t/k) - M(u/k)$, $2 \le k \le K$; for $K = 2$ and $\lambda = 5/4$, the only channel above $5u/12$ is the increment across $(u/2, t/2]$. The one-step recursion does not by itself give an iterable maximum principle, and a termwise absolute-value contraction gives no nontrivial gain. The exact intra-excursion formulation, the scale cascade, a persistence extraction lemma, and a ballistic counterexample to the relaxed divisor-sum version are given below.

Extreme values. If a normalized excursion statistic has a uniform tail $\Pr(Z > z) \le Ce^{-cz^p}$, the counting bounds force the maximum over a family with at most $L(X)$ members starting below X to be of order $(\log L(X))^{1/p}$ in probability, with an almost sure lim sup constant $c^{-1/p}$ (Theorems 8.61 and 8.63). Each level of lacunarity removes one logarithm from the extreme scale: $\sqrt{\log X}$, $\sqrt{\log\log X}$, $\sqrt{\log\log\log X}$.

The remaining sections contain exact identities for the skeleton and its area curves (Sections 2–3), exact correlation identities and sharp bounds inside an excursion (Section 4), the relation with ζ (Section 5), area bounds and global criteria equivalent to RH (Sections 6–7), the exact bilinear form of M on the square range of an excursion and the Mellin mass of an excursion (Section 8), and the decomposition of M by large prime factors, globally and inside an excursion (Sections 9–11). Section 13 records numerical data on the excursion census.

Conventions. $\rho = \beta + i\gamma$ runs over nontrivial zeros of ζ,

$$\Theta = \sup\{\Re\rho\}, \qquad \gamma_1 = 14.134725\ldots$$

is the ordinate of the lowest zero, and RH means $\Theta = 1/2$. Littlewood's criterion [18, §14.25] states that RH holds if and only if $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$ for every $\varepsilon > 0$. Statements labelled *Heuristic* are not used in any proof.

2. Excursions and the regularized skeleton

Let μ be the Möbius function and

$$M(x) = \sum_{n\le x} \mu(n), \qquad x \ge 0, \tag{2.1}$$

piecewise constant and right-continuous, with jumps $\mu(n) \in \{-1, 0, 1\}$ at the integers. The segmentation of M by zero runs and the record of its extrema are standard computational objects; Hurst [6] computed $M(x)$ for all $x \le 10^{16}$. The area and quadratic functionals of lattice excursions are classical [17, 14, 16, 8, 2, 1]; here they are applied to the deterministic Möbius walk.

Definition 2.1 (Excursions and zero runs)**.** An *excursion* is an interval

$$I_\nu = (x_{0,\nu}, x_{1,\nu}]$$

with integer endpoints, $M(x_{0,\nu}) = M(x_{1,\nu}) = 0$, $M(x_{0,\nu}+1) = \sigma_\nu \in \{\pm 1\}$, and $\sigma_\nu M(n) \ge 1$ for every integer $x_{0,\nu} < n < x_{1,\nu}$. Thus $x_{1,\nu}$ is the first return to zero after the departure from $x_{0,\nu}$.

Consecutive excursions may be separated by a *zero run* $[x_{1,\nu}, x_{0,\nu+1}]$, a maximal set of consecutive integers on which $M = 0$; it may consist of a single point. The first excursion is $(0, 2]$. Since M takes values of both signs of size $\gg \sqrt{x}$ infinitely often (Remark 6.6) and has unit steps, it has infinitely many zeros and every excursion is finite. We also write $u_\nu = x_{0,\nu}$ and $v_\nu = x_{1,\nu}$.

Put

$$\ell_\nu = x_{1,\nu} - x_{0,\nu}, \qquad H_\nu = \max_{x_{0,\nu}<n<x_{1,\nu}} |M(n)|,$$

$$\tau_\nu = \min\{n \in I_\nu : |M(n)| = H_\nu\}, \qquad a_\nu = \tau_\nu - x_{0,\nu}, \qquad b_\nu = x_{1,\nu} - \tau_\nu.$$

Every excursion has $H_\nu \geq 1$ and $\ell_\nu \geq 2$.

Lemma 2.2 (Height and step count)**.** *For every excursion,*

$$H_\nu \leq \min(a_\nu, b_\nu) \leq \frac{\ell_\nu}{2}. \tag{2.2}$$

Proof. The steps of M are 0 or ± 1, so at least H_ν steps are needed to reach height H_ν from zero and at least H_ν more to return to zero. □

Definition 2.3 (Crossing and touching zero runs)**.** The zero run $[x_{1,\nu}, x_{0,\nu+1}]$ between consecutive excursions is a *crossing run* if $\sigma_{\nu+1} = -\sigma_\nu$ and a *touching run* if $\sigma_{\nu+1} = \sigma_\nu$.

Definition 2.4 (Regularized skeleton)**.** The *skeleton* M_{s} is the continuous, piecewise linear function defined on every excursion by

$$M_{\mathrm{s}}(t) = \begin{cases} \sigma_\nu \sqrt{H_\nu}\, \dfrac{t - x_{0,\nu}}{a_\nu}, & x_{0,\nu} \leq t \leq \tau_\nu, \\[2ex] \sigma_\nu \sqrt{H_\nu}\, \dfrac{x_{1,\nu} - t}{b_\nu}, & \tau_\nu \leq t \leq x_{1,\nu}, \end{cases} \tag{2.3}$$

and by $M_{\mathrm{s}} \equiv 0$ on zero runs. The *unregularized skeleton* M_{sk} is defined in the same way with peak $\sigma_\nu H_\nu$.

Definition 2.5 (Terminal block of a truncated computation)**.** Let $N \geq 2$ and $u = \max\{n \leq N : M(n) = 0\}$. If $u = N$ there is no terminal block. Otherwise $M(n) \neq 0$ for $u < n \leq N$, and since M has steps $0, \pm 1$ all these values have the sign of $M(N)$. Put

$$\sigma_{\mathrm{term}} = \operatorname{sgn} M(N), \qquad H_{\mathrm{term}} = \max_{u<n\leq N} |M(n)|, \qquad \tau_{\mathrm{term}} = \min\{n \in (u, N] : |M(n)| = H_{\mathrm{term}}\}.$$

On $[u, N]$ the regularized skeleton is the segment from $(u, 0)$ to $(\tau_{\mathrm{term}}, \sigma_{\mathrm{term}}\sqrt{H_{\mathrm{term}}})$ followed, if $\tau_{\mathrm{term}} < N$, by the segment ending at $(N, \sigma_{\mathrm{term}}\sqrt{|M(N)|})$. Its signed area is

$$\Delta\mathcal{A}(u, N] = \frac{\sigma_{\mathrm{term}}}{2}\Big[\sqrt{H_{\mathrm{term}}}(N - u) + \sqrt{|M(N)|}(N - \tau_{\mathrm{term}})\Big].$$

The terminal skeleton is provisional: after the next return to zero the completed excursion acquires a different descending side. Independently of it, for real x with last zero $u(x) \leq \lfloor x \rfloor$ define

$$\beta_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} |M(n)|, \qquad Q_{\mathrm{term}}(x) := \sum_{u(x)<n\leq\lfloor x\rfloor} M(n)^2, \qquad L_x := \lfloor x \rfloor - u(x).$$

These depend only on M.

Definition 2.6 (The three curves)**.** With $M_{\mathrm{s}}^{\pm} = \max(\pm M_{\mathrm{s}}, 0)$,

$$\mathcal{A}_+(x) = \int_0^x M_{\mathrm{s}}^+(t)\, dt, \qquad \mathcal{A}_-(x) = -\int_0^x M_{\mathrm{s}}^-(t)\, dt, \qquad \mathcal{A}(x) = \int_0^x M_{\mathrm{s}}(t)\, dt. \tag{2.4}$$

$\mathcal{A}_+$ is nondecreasing, $\mathcal{A}_-$ is nonincreasing, and both are constant on zero runs and on excursions of the opposite sign.

Remark 2.7 (The family of g-skeletons). For any $g : \mathbb{N}_0 \to [0,\infty)$ with $g(0) = 0$ and $g(h) > 0$ for $h \geq 1$, one may replace the apex height $\sqrt{H}$ by $g(H)$. The signed area increment becomes $\frac{1}{2}\sigma_\nu g(H_\nu)\ell_\nu$. The correlation identities of Section 4 concern the Möbius increments and do not depend on g. The choice $g(H) = \sqrt{H}$ is a normalization.

3. Exact formulas

Lemma 3.1 (Increment per excursion). *For every complete excursion,*

$$\mathcal{A}(x_{1,\nu}) - \mathcal{A}(x_{0,\nu}) = \sigma_\nu \frac{\sqrt{H_\nu}\,\ell_\nu}{2}, \qquad \sum_{n\in I_\nu} M_{\mathrm{s}}(n) = \sigma_\nu \sqrt{H_\nu}\frac{\ell_\nu}{2}. \tag{3.1}$$

Proof. The two pieces are triangles of height $\sqrt{H_\nu}$ and bases a_ν, b_ν. For the discrete sum, the ascending values at $j = 1,\dots,a_\nu$ sum to $\sqrt{H_\nu}\sum_{j\leq a_\nu} j/a_\nu = \sqrt{H_\nu}(a_\nu+1)/2$; the descending values at $\tau_\nu + r$, $r = 1,\dots,b_\nu$, sum to $\sqrt{H_\nu}\sum_{r\leq b_\nu}(b_\nu - r)/b_\nu = \sqrt{H_\nu}(b_\nu - 1)/2$. □

Proposition 3.2 (Explicit form of the three curves). *Let $x \in I_\nu$. Then*

$$\mathcal{A}(x) = \begin{cases} \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\dfrac{(x - x_{0,\nu})^2}{2a_\nu}, & x \leq \tau_\nu, \\[2ex] \mathcal{A}(x_{0,\nu}) + \sigma_\nu\sqrt{H_\nu}\left(\dfrac{a_\nu}{2} + (x-\tau_\nu) - \dfrac{(x-\tau_\nu)^2}{2b_\nu}\right), & x \geq \tau_\nu. \end{cases} \tag{3.2}$$

At excursion endpoints,

$$\mathcal{A}(x_{1,k}) = \sum_{\nu\leq k}\sigma_\nu\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_+(x_{1,k}) = \sum_{\substack{\nu\leq k\\ \sigma_\nu=+1}}\frac{\sqrt{H_\nu}\ell_\nu}{2}, \qquad \mathcal{A}_-(x_{1,k}) = -\sum_{\substack{\nu\leq k\\ \sigma_\nu=-1}}\frac{\sqrt{H_\nu}\ell_\nu}{2}. \tag{3.3}$$

The curves are piecewise quadratic and C^1.

Proof. Integrate (2.3); the endpoint formulas follow from Lemma 3.1. □

A point c is a *strict inflection point* of a function if there exists $\delta > 0$ such that the function is strictly convex on one side of c and strictly concave on the other, within distance δ. The second derivative need not exist at c.

Lemma 3.3 (Critical geometry of $\mathcal{A}$). *Let $R = [p,q]$ be the zero run between an excursion of sign σ_- ending at p and the next excursion of sign σ_+ starting at q.*

(i) *Every apex τ_ν is a strict inflection point.*
(ii) *If $\sigma_+ = -\sigma_-$, then R is an extremal plateau; it is a strict local extremum exactly when $p = q$.*
(iii) *If $\sigma_+ = \sigma_- =: \sigma$, then $\mathcal{A}$ has no local extremum on R. Its concavity has sign $-\sigma$ immediately to the left and $+\sigma$ immediately to the right. If $p = q$ the point is a strict inflection; if $p < q$, no point of R is a strict inflection, and the change of concavity is carried by the whole plateau.*
(iv) *Strict local extrema occur exactly at isolated crossing zeros, and plateaus of extrema exactly at crossing runs of positive length; every point of such a plateau is a non-strict local extremum.*

Proof. On the descending side of the left excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_-$ and $\operatorname{sgn}\mathcal{A}'' = -\sigma_-$. On the ascending side of the right excursion, $\operatorname{sgn}\mathcal{A}' = \sigma_+$ and $\operatorname{sgn}\mathcal{A}'' = \sigma_+$. On R, $\mathcal{A}' = 0$. The assertions follow, and at every apex the sign of $\mathcal{A}''$ reverses. □

4. Internal correlations of an excursion

For a real block $x_1, \dots, x_L$ put

$$S_t = \sum_{j=1}^{t} x_j, \qquad C_h = \sum_{j=1}^{L-h} x_j x_{j+h}. \tag{4.1}$$

Lemma 4.1 (Triangular identity). *For every real finite sequence,*

$$\sum_{t=1}^{L-1} S_t^2 + \sum_{h=1}^{L-1} hC_h = S_L \sum_{t=1}^{L-1} S_t. \tag{4.2}$$

Proof. Expanding the partial sums,

$$\sum_{t=1}^{L} S_t^2 = \sum_{i,j=1}^{L} x_i x_j \big(L + 1 - \max(i,j)\big).$$

Write $\max(i,j) = \frac{1}{2}(i + j + |i-j|)$. The part containing $|i-j|$ is $\frac{1}{2}\sum_{i,j} x_i x_j |i-j| = \sum_{h=1}^{L-1} hC_h$. The remaining part is $(L+1)S_L^2 - S_L \sum_i i x_i$, and $\sum_i i x_i = LS_L - \sum_{t=1}^{L-1} S_t$. Hence $\sum_{t=1}^{L} S_t^2 = S_L^2 + S_L \sum_{t<L} S_t - \sum_h hC_h$; subtract S_L^2. □

Corollary 4.2 (Bridge case). *If $S_L = 0$, then*

$$\sum_{t=1}^{L-1} S_t^2 = -\sum_{h=1}^{L-1} hC_h. \tag{4.3}$$

Remark 4.3 (Hermitian form). For complex x_j and $C_h = \sum_{j=1}^{L-h} x_j \overline{x_{j+h}}$, the same computation gives

$$\sum_{t=1}^{L-1} |S_t|^2 + \sum_{h=1}^{L-1} h\Re C_h = \Re\left(\overline{S_L} \sum_{t=1}^{L-1} S_t \right).$$

For a complete excursion orient the increments by

$$x_j = \sigma_\nu \mu(x_{0,\nu} + j), \qquad S_t = |M(x_{0,\nu} + t)|. \tag{4.4}$$

Then $S_t > 0$ for $1 \le t < \ell_\nu$ and $S_{\ell_\nu} = 0$. Put

$$Q_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t^2, \qquad C_h^{(\nu)} = \sum_{j=1}^{\ell_\nu - h} x_j x_{j+h}, \qquad \text{so that} \qquad Q_\nu = -\sum_{h=1}^{\ell_\nu - 1} hC_h^{(\nu)}. \tag{4.5}$$

Lemma 4.4 (Zeroth moment and extreme lag). *With $q_\nu = \sum_j x_j^2$,*

$$\sum_{h=1}^{\ell_\nu - 1} C_h^{(\nu)} = -\frac{q_\nu}{2}, \qquad C_{\ell_\nu - 1}^{(\nu)} = -1, \tag{4.6}$$

and q_ν is even and at least 2.

Proof. $0 = (\sum_j x_j)^2 = q_\nu + 2\sum_h C_h^{(\nu)}$. The first oriented step is $+1$; immediately before the first return the path is at height 1, so the last step is -1. The bridge condition forces equally many positive and negative active steps. □

Proposition 4.5 (Quadratic decomposition by excursions). *For real $x \ge 1$ let $u(x)$ and L_x be as in Definition 2.5, and for $1 \le j \le L_x$ put $y_j = \mu(u(x) + j)$ and $C_h^{\mathrm{term}} = \sum_{j=1}^{L_x - h} y_j y_{j+h}$. Then*

$$\sum_{n \le x} M(n)^2 = \sum_{\nu : x_{1,\nu} \le u(x)} Q_\nu + Q_{\mathrm{term}}(x), \tag{4.7}$$

$$Q_{\mathrm{term}}(x) = M(\lfloor x \rfloor)^2 + M(\lfloor x \rfloor) \sum_{t=1}^{L_x - 1} M(u(x)+t) - \sum_{h=1}^{L_x-1} h C_h^{\mathrm{term}}, \tag{4.8}$$

with $Q_{\mathrm{term}}(x) = 0$ *if* $L_x = 0$. *At a zero* x *of* M,

$$\sum_{n \le x} M(n)^2 = \sum_{x_{1,\nu} \le x} Q_\nu = - \sum_{x_{1,\nu} \le x} \sum_h h C_h^{(\nu)}. \tag{4.9}$$

Proof. Every integer up to $u(x)$ lies either in a zero run or in a unique excursion interior. Apply Lemma 4.1 to the terminal block and add its final term $S_{L_x}^2$. □

Thus the quadratic sum sees only lag correlations internal to excursions: pairs of Möbius increments in different completed excursions do not contribute.

Let

$$\beta_\nu = \sum_{t=1}^{\ell_\nu - 1} S_t = \sum_{n \in I_\nu} |M(n)|. \tag{4.10}$$

Cauchy–Schwarz gives

$$\beta_\nu^2 \le (\ell_\nu - 1) Q_\nu, \tag{4.11}$$

with equality exactly for a flat excursion of height 1. To separate support from sign geometry, let $P_h^{(\nu)} = \#\{j : x_j x_{j+h} \neq 0\}$ and let $D_h^{(\nu)}$ count the opposite-sign pairs among them. Then

$$C_h^{(\nu)} = P_h^{(\nu)} - 2D_h^{(\nu)}, \qquad Q_\nu = \sum_{h=1}^{\ell_\nu - 1} h\big(2D_h^{(\nu)} - P_h^{(\nu)}\big). \tag{4.12}$$

Proposition 4.6 (Sharp bounds at fixed length and height)**.** *Let* $S_0 = S_L = 0$, $S_t \ge 1$ *for* $0 < t < L$, $S_t - S_{t-1} \in \{-1, 0, 1\}$ *and* $H = \max S_t$. *Then* $L \ge 2H$ *and*

$$H^2 + L - 2H \le \beta \le H(L - H), \tag{4.13}$$

$$\frac{2H^3 + H}{3} + L - 2H \le Q \le \frac{2H^3 + H}{3} + (L - 2H)H^2. \tag{4.14}$$

All four bounds are attained. In particular

$$Q \ge \frac{2H^3 + H}{3}, \qquad Q \ge L - 1. \tag{4.15}$$

Proof. Before and after the first visit to H the path visits each level $1, \dots, H-1$. Together with one visit to H, these $2H - 1$ designated interior times contribute $2(1 + \cdots + (H-1)) + H = H^2$ to β and $2\sum_{j<H} j^2 + H^2 = (2H^3 + H)/3$ to Q. The remaining $L - 2H$ interior times lie at heights between 1 and H; placing them all at height 1, or all at height H, attains the bounds. □

Corollary 4.7 (Sharp fixed-length bounds)**.** *For* $L \ge 2$,

$$\beta \le \left\lfloor \frac{L^2}{4} \right\rfloor, \qquad Q \le \begin{cases} L(L^2+2)/12, & L \text{ even}, \\ L(L^2-1)/12, & L \text{ odd}, \end{cases}$$

and in particular

$$Q \le \frac{L^3 + 2L}{12}. \tag{4.16}$$

Both bounds are attained by the tent path, with one horizontal step at maximal height when L *is odd.*

Proof. Use $S_t \le \min(t, L - t)$ and sum this bound and its square. □

5. Where ζ enters

The black curve depends on M nonlinearly, through $\sqrt{H_\nu}$ and through the selection of $x_{0,\nu}, \tau_\nu, x_{1,\nu}$, and no closed expression of its Mellin transform in terms of ζ is known. The nodes $x_{0,\nu}, x_{1,\nu}$ are zeros of M. If RH holds and all zeros are simple, there is a sequence $T_k \to \infty$ such that, for non-integer x,

$$M(x) = \lim_{k\to\infty} \sum_{|\gamma|<T_k} \frac{x^\rho}{\rho\,\zeta'(\rho)} - 2 + \sum_{n\ge 1} \frac{(-1)^{n-1}(2\pi/x)^{2n}}{(2n)!\,n\,\zeta(2n+1)} \tag{5.1}$$

[18, §14.27]. For $\widetilde{A}(x) := \int_0^x M(t)\,dt$ one has, for $\Re s > 1$,

$$\int_1^\infty \widetilde{A}(x) x^{-s-2}\,dx = \frac{1}{s(s+1)\zeta(s)}. \tag{5.2}$$

Shifting the contour formally, the residues are

$$\sum_\rho \frac{x^{\rho+1}}{\rho(\rho+1)\zeta'(\rho)} - 2x + 12 + (\text{residues at the trivial zeros}), \tag{5.3}$$

where $x/\zeta(0) = -2x$ is the residue at $s = 0$ and $1/((-1)\zeta(-1)) = 12$ the residue at $s = -1$. The factor $(\rho+1)^{-1}$ improves convergence, but RH and simplicity alone do not control $1/\zeta'(\rho)$ well enough to turn (5.3) into an identity with an untruncated sum; results of this type require additional input of Gonek–Hejhal type [13]. The area of M_{sk} differs from $\widetilde{A}$ by explicit local terms.

Proposition 5.1 (Dipole identity)**.** *For each excursion let its* legs *be the integer sets $\lambda = (u_\nu, \tau_\nu] \cap \mathbb{Z}$ and $(\tau_\nu, v_\nu] \cap \mathbb{Z}$, with midpoints $c_\lambda = (u_\nu + \tau_\nu)/2$ and $(\tau_\nu + v_\nu)/2$. Then for every zero v_k,*

$$\int_0^{v_k} M_{\mathrm{sk}}(t)\,dt = \widetilde{A}(v_k) + \sum_{\lambda\subset(0,v_k]} \sum_{m\in\lambda} \mu(m)\,(m - c_\lambda). \tag{5.4}$$

Proof. Since M is constant on each $[n, n+1)$, $\int_a^b M(t)\,dt = \sum_{n=a}^{b-1} M(n)$ for integers $a < b$. On an ascent leg, $\int_u^\tau M_{\mathrm{sk}} = \sigma H a/2$ and $\int_u^\tau M = \sum_{m=u+1}^{\tau} \mu(m)(\tau - m)$, while $\sigma H = \sum_{m=u+1}^{\tau} \mu(m)$; the difference is $\sum_m \mu(m)(a/2 - \tau + m) = \sum_m \mu(m)(m - c_\lambda)$. On a descent leg, $\int_\tau^v M_{\mathrm{sk}} = \sigma H b/2$ and $\int_\tau^v M = b\,\sigma H + \sum_{m=\tau+1}^{v} \mu(m)(v - m)$ with $\sum_{m=\tau+1}^{v} \mu(m) = -\sigma H$; the difference is $\sum_m \mu(m)(m - (v - b/2))$, and $v - b/2 = (\tau + v)/2 = c_\lambda$. On zero runs both integrands vanish. Sum over legs. □

6. Bounds for the area curves

Lemma 6.1 (Transfer)**.** *Let $x \ge 1$ and let $x^* \ge x$ be a zero of M (in a computation truncated at $N \ge x$, take $x^* = N$ and the terminal skeleton of Definition 2.5). If $|M(n)| \le c\,n^{1/2}$ for all $1 \le n \le x^*$ (so necessarily $c \ge 1$), then*

$$|M_{\mathrm{s}}(t)| \le c^{1/2} t^{1/4} \quad (0 \le t \le x), \qquad \mathcal{A}_+(x) + |\mathcal{A}_-(x)| \le \tfrac{4}{5}\, c^{1/2} x^{5/4}. \tag{6.1}$$

Proof. For $t \ge \tau_\nu$, $|M_{\mathrm{s}}(t)| \le \sqrt{H_\nu} \le c^{1/2}\tau_\nu^{1/4} \le c^{1/2} t^{1/4}$. For $u_\nu \le t \le \tau_\nu$ the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$ lies below the chord of $c^{1/2}t^{1/4}$ over $[u_\nu, \tau_\nu]$, whose endpoints $c^{1/2}u_\nu^{1/4} \ge 0$ and $c^{1/2}\tau_\nu^{1/4} \ge \sqrt{H_\nu}$ (as $\tau_\nu \le x^*$) are at least as high; that chord lies below $c^{1/2}t^{1/4}$ by concavity. Integrate. □

Lemma 6.2 (A concave majorant)**.** *Let $\phi(y) = (\log y)^{3/5}(\log\log y)^{-1/5}$ and, for $\kappa > 0$, $F(y) = y^{1/2} e^{-\kappa\phi(y)}$. There is $y_0 = y_0(\kappa)$ such that F is increasing and concave on $[y_0, \infty)$.*

Proof. With $y = e^s$ and $g(s) = s^{3/5}(\log s)^{-1/5}$, one has $F = \exp(s/2 - \kappa g(s))$. Put $h(s) = \frac{1}{2} - \kappa g'(s)$. Then

$$yF'(y) = F\,h(s), \qquad y^2 F''(y) = F\big(h(s)^2 - h(s) - \kappa g''(s)\big).$$

Since $g'(s) \to 0$ and $g''(s) \to 0$, $h(s) \to \frac{1}{2}$ and $h^2 - h - \kappa g'' \to -\frac{1}{4}$. Hence $F' > 0$ and $F'' < 0$ for large y. □

Corollary 6.3. *From the Vinogradov–Korobov bound* [19]

$$M(x) \ll x \exp\bigl(-c\,\phi(x)\bigr) \tag{6.2}$$

one obtains

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll x^{3/2} \exp\bigl(-\tfrac{c}{2}\,\phi(x)\bigr). \tag{6.3}$$

Proof. Let K be such that $|M(y)| \le K y\, e^{-c\phi(y)}$ for $y \ge y_1$, and let F be as in Lemma 6.2 with $\kappa = c/2$, so that $\sqrt{|M(y)|} \le K^{1/2} F(y)$ for $y \ge y_1$. Only finitely many excursions start below $Y_0 := \max(y_0, y_1)$; they are finite, so they contribute $O(1)$ to the area. Consider an excursion with $u_\nu \ge Y_0$. Its apex satisfies $\sqrt{H_\nu} = \sqrt{|M(\tau_\nu)|} \le K^{1/2} F(\tau_\nu)$; the bound is taken at the apex τ_ν, which lies in the excursion, and not at its future endpoint v_ν. For $t \ge \tau_\nu$ monotonicity of F gives $|M_{\mathrm{s}}(t)| \le \sqrt{H_\nu} \le K^{1/2} F(t)$. For $u_\nu \le t \le \tau_\nu$, $|M_{\mathrm{s}}(t)|$ is the chord from $(u_\nu, 0)$ to $(\tau_\nu, \sqrt{H_\nu})$. This chord lies below the chord from $(u_\nu, K^{1/2}F(u_\nu))$ to $(\tau_\nu, K^{1/2}F(\tau_\nu))$, which has both endpoints at least as high, and the latter lies below $K^{1/2}F$ on $[u_\nu, \tau_\nu] \subset [y_0, \infty)$ by concavity. Hence $|M_{\mathrm{s}}(t)| \le K^{1/2} F(t)$ for all $t \ge Y_0$ outside the finitely many early excursions, and since F is increasing,

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \le O(1) + K^{1/2} \int_{Y_0}^{x} F(t)\,dt \le O(1) + K^{1/2} x F(x) = O(1) + K^{1/2} x^{3/2} e^{-\frac{c}{2}\phi(x)}. \quad \square$$

Corollary 6.4. *If RH holds, then*

$$\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \ll_\varepsilon x^{5/4+\varepsilon}. \tag{6.4}$$

Consequently, if $\mathcal{A}_+(x) + |\mathcal{A}_-(x)| \gg x^{5/4+\delta}$ along an unbounded sequence for some fixed $\delta > 0$, then RH is false.

Remark 6.5 (Endpoint-indexed bounds). From $\ell_\nu \ge 2H_\nu$ and $\sqrt{H_\nu}\ell_\nu \ll v_\nu^{5/4+\varepsilon}$ one deduces only $H_\nu \ll v_\nu^{5/6+O(\varepsilon)}$. This bound is indexed by the *future* endpoint v_ν of the excursion; it gives no bound for $M(x)$ in terms of x at interior points, and hence no zero-free half-plane. Whether (6.4) is equivalent to RH is open.

Remark 6.6 (Oscillation of the heights). The Mertens conjecture is false [15, 10]; the current bounds are due to Hurst [6], who proves $\limsup M(x)/\sqrt{x} > 1.826054$ and $\liminf M(x)/\sqrt{x} < -1.837625$. The excursion containing a point n with $M(n) > 1.826054\sqrt{n}$ has $u_\nu < n$, so $H_\nu > 1.826054\, u_\nu^{1/2}$ for infinitely many positive excursions, and similarly with 1.837625 for negative ones. A lower bound indexed by v_ν does not follow, since v_ν may be much larger than n.

7. Global criteria equivalent to RH

The Mellin identity and the global L^1/L^2 equivalences below are consequences of Littlewood's criterion and partial summation; the excursion decompositions and pruning lemmas repackage them in terms of excursions. For $\Re s > 1$,

$$\int_1^\infty M(x) x^{-s-1}\,dx = \frac{1}{s\,\zeta(s)}. \tag{7.1}$$

If $\int_1^\infty |M(x)| x^{-\sigma-1} dx < \infty$ for every $\sigma > 1/2$, the left side continues analytically to $\Re s > 1/2$, so ζ has no zeros there.

Proposition 7.1 (L^1 criterion)**.** *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{v_\nu \le x} \beta_\nu + \beta_{\mathrm{term}}(x) = \sum_{n \le x} |M(n)| \ll x^{3/2+\varepsilon}. \tag{7.2}$$

Proof. RH gives $M(x) \ll_\varepsilon x^{1/2+\varepsilon}$. Conversely, for $\sigma > 1/2$ choose $0 < \varepsilon < \sigma - 1/2$; partial summation gives $\sum_n |M(n)| n^{-\sigma-1} < \infty$, and since M is constant on $[n, n+1)$ the remark after (7.1) applies. □

Lemma 7.2 (Pruning). *$\beta_\nu \le H_\nu(\ell_\nu - 1) \le \ell_\nu^2/2$, hence*

$$\sum_{v_\nu \le x,\ \ell_\nu \le L_0} \beta_\nu \le \frac{xL_0}{2}. \tag{7.3}$$

Corollary 7.3 (Reduction to long excursions). *For fixed $\delta \ge 0$ and $L_0 = x^{1/2-\delta}$, RH holds if and only if*

$$\sum_{\substack{v_\nu \le x \\ \ell_\nu > L_0}} \beta_\nu + \mathbf{1}_{\{L_x > L_0\}} \beta_{\mathrm{term}}(x) \ll x^{3/2+\varepsilon} \quad \text{for every } \varepsilon > 0. \tag{7.4}$$

At most x/L_0 long excursions are involved.

Remark 7.4 (The $\sqrt{x}$ frontier). The cut may depend on ε: with $L_0 = x^{1/2+\varepsilon/2}$ the discarded part is $\le x^{3/2+\varepsilon/2}/2$. The frontier of pruning is therefore $x^{1/2+o(1)}$; what requires new information is a cut x^θ with a fixed $\theta > 1/2$. For $\ell_\nu \asymp x$ the trivial bound gives $\beta_\nu \ll x^2$ against a budget $x^{3/2+\varepsilon}$, and there are only $O(1)$ such excursions.

Heuristic 7.5 (Renewal tail). If excursion lengths behave like those of a recurrent random walk,

$$\#\{v_\nu \le x :\ \ell_\nu > L\} \asymp \sqrt{x}\, L^{-1/2}, \tag{7.5}$$

and with $\beta_\nu \asymp \ell_\nu^{3/2}$ the dyadic range $[L, 2L]$ contributes $\asymp \sqrt{x}\, L$ to (7.2), so the terminal scales $L \asymp x$ dominate the L^1 sum.

Theorem 7.6 (Correlation criterion). *RH holds if and only if, for every $\varepsilon > 0$,*

$$\sum_{n \le x} M(n)^2 = \sum_{v_\nu \le x} Q_\nu + Q_{\mathrm{term}}(x) = -\sum_{v_\nu \le x} \sum_h h C_h^{(\nu)} + Q_{\mathrm{term}}(x) \ll x^{2+\varepsilon}. \tag{7.6}$$

Proof. Under RH, $M(n)^2 \ll_\varepsilon n^{1+\varepsilon}$. Conversely, by Cauchy–Schwarz,

$$\sum_{n \le x} |M(n)| \le x^{1/2} \Big(\sum_{n \le x} M(n)^2 \Big)^{1/2} \ll x^{3/2+\varepsilon/2};$$

apply Proposition 7.1. □

Corollary 7.7 (Local sufficient conditions). *Suppose that for every $\varepsilon > 0$ there is C_ε such that*

$$Q_\nu \le C_\varepsilon\, \ell_\nu^{2+\varepsilon} \quad \text{for every complete excursion with } \ell_\nu > v_\nu^{1/2}, \tag{7.7}$$

and $Q_{\mathrm{term}}(x) \le C_\varepsilon L_x^{2+\varepsilon}$ whenever $L_x > x^{1/2}$. Then RH holds. In particular it suffices that (7.7) hold for every complete excursion.

Proof. Split the completed excursions at $\ell_\nu = x^{1/2}$. If $\ell_\nu \le x^{1/2}$, then (4.16) gives $Q_\nu \le \ell_\nu x/12 + \ell_\nu/6$, and disjointness bounds the total by $x^2/12 + x/6$. If $\ell_\nu > x^{1/2}$ then $\ell_\nu > v_\nu^{1/2}$ since $v_\nu \le x$, and the hypothesis bounds the total by $C_\varepsilon \sum \ell_\nu^{2+\varepsilon} \le C_\varepsilon (\sum \ell_\nu)^{2+\varepsilon} \le C_\varepsilon x^{2+\varepsilon}$. The terminal block is not a complete excursion, so (7.7) does not apply to it; it is bounded by (4.16) if $L_x \le x^{1/2}$ and by the terminal hypothesis otherwise. Apply Theorem 7.6. □

Heuristic 7.8 (Ballistic excursions). In a model where, after a zero of M, the next values of μ behave locally at random with squarefree density $6/\pi^2$, the pattern "k non-zero steps of one sign followed by k of the other" follows a given zero with probability $\ge c_0^{-k}$. If excursions with $H_\nu = k$ and $\ell_\nu \asymp k$ occur for unbounded k, then by (4.15) $Q_\nu/\ell_\nu^{2+\varepsilon} \gg k^{1-\varepsilon} \to \infty$, so the unpruned version

of (7.7) fails for every $\varepsilon \in (0,1)$. Such excursions have $\ell_\nu \ll v_\nu^{1/2}$ and do not affect the pruned condition (7.7); this is why the pruning in Corollary 7.7 is needed.

Theorem 7.9 (Energy criterion). *With u the last zero $\le x$ and $H_{\mathrm{run}}(x) = \max_{u<n\le x}|M(n)|$, define*

$$\mathcal{E}(x) := \frac{1}{4}\Big(\sum_{v_\nu \le x} H_\nu \ell_\nu + (x-u)H_{\mathrm{run}}(x)\Big). \tag{7.8}$$

For a complete excursion let J be the chord joining $(u_\nu, \mathcal{A}(u_\nu))$ to $(v_\nu, \mathcal{A}(v_\nu))$. By (3.1), $J' = \sigma_\nu\sqrt{H_\nu}/2$, so $\frac{1}{4}H_\nu\ell_\nu = \int_{I_\nu}(J')^2$ is the Dirichlet energy of the chord. Then

$$\sum_{n\le x}|M(n)| \le 4\mathcal{E}(x) \le x\max_{n\le x}|M(n)|, \tag{7.9}$$

and RH holds if and only if $\mathcal{E}(x) \ll x^{3/2+\varepsilon}$ for every $\varepsilon > 0$.

Proof. $\beta_\nu \le H_\nu\ell_\nu$, $\beta_{\mathrm{term}}(x) \le (x-u)H_{\mathrm{run}}(x)$ and $\sum \ell_\nu + (x-u) \le x$; apply Proposition 7.1. □

8. Lacunary excursion families and horizon-localized criteria

8.1. **Geometric counts.** The lemmas of this subsection use only that excursions are disjoint integer intervals ordered by their starting points, the next one starting at or after the end of the previous one. They therefore hold verbatim for every sequence of disjoint integer intervals, deterministic or random, and in particular pathwise for any stochastic model of excursions (Section 8.6). For a family $\mathcal{F}$ of excursions and $X \ge 2$ put

$$E_{\mathcal{F}}(X) := \#\{\nu \in \mathcal{F} : x_{0,\nu} \le X\}.$$

For a function h on $[0,\infty)$ with $h(x) \ge x$, the family of excursions *crossing the horizon* h is

$$\mathcal{F}_h := \{\nu : x_{1,\nu} > h(x_{0,\nu})\}.$$

For $A > 1$ the excursion I_ν is *A-exceptional* if $x_{1,\nu} > x_{0,\nu}^A$, i.e. $\nu \in \mathcal{F}_{x^A}$, and *$A$-ordinary* otherwise; for $A = 2$ we say *square-exceptional* and *square-ordinary*. For $\lambda > 1$ it is *λ-long* if $x_{1,\nu} > \lambda x_{0,\nu}$. We write $E_A = E_{\mathcal{F}_{x^A}}$, $E_\square = E_2$ and $E_\lambda = E_{\mathcal{F}_{\lambda x}}$; the context separates E_A (A an exponent) from E_λ (λ a factor).

Lemma 8.1 (Power lacunarity). *For every $A > 1$ and $X \ge 2$,*

$$E_A(X) \le 2 + \frac{1}{\log A}\log\frac{\log X}{\log 2}. \tag{8.1}$$

In particular

$$E_\square(X) \le 2 + \log_2\frac{\log X}{\log 2}, \qquad E_\square(X) \le 4\log\log X \quad (X \ge e^e). \tag{8.2}$$

Proof. No excursion starts at 1, since $M(1) = 1$. List the A-exceptional starts ≥ 2 as $2 \le a_1 < a_2 < \cdots$. The next excursion starts at or after the end of the preceding one, so $a_{j+1} \ge x_{1,j} > a_j^A$ and $\log a_j \ge A^{j-1}\log 2$. If $a_j \le X$ then $j \le 1 + \log(\log X/\log 2)/\log A$; add 1 for the excursion $(0,2]$. For the second bound in (8.2), $\log_2(\log X/\log 2) = (\log\log X - \log\log 2)/\log 2 \le 1.443\log\log X + 0.529$. □

The next lemma is the partition of the square horizon $[x_0, x_0^2] = [x_0, cx_0^2] \cup [cx_0^2, x_0^2]$: the upper part is accessible only to a doubly lacunary family, for every fixed $c > 0$.

Lemma 8.2 (Fractional square horizon)**.** *For $0 < c \le 1$ and $X \ge 2$,*

$$E_{\mathcal{F}_{cx^2}}(X) = \#\{\nu : x_{0,\nu} \le X,\ x_{1,\nu} > cx_{0,\nu}^2\} \le \frac{2}{c} + 2 + \log_2 \frac{\log X}{\log 2}. \tag{8.3}$$

Consequently, the range $cx_0^2 < t \le x_0^2$ of the excursions starting at most X is visited by at most $O_c(\log\log X)$ excursions.

Proof. At most $\lceil 2/c \rceil \le 2/c + 1$ members start below $2/c$. List the others as $a_1 < a_2 < \cdots$ and put $b_j = ca_j \ge 2$. Then $a_{j+1} \ge x_{1,j} > ca_j^2$, so

$$b_{j+1} = ca_{j+1} > c^2 a_j^2 = b_j^2,$$

and $\log b_j \ge 2^{j-1} \log 2$. Since $b_j \le cX \le X$, the number of such j is at most $1 + \log_2(\log X/\log 2)$. An excursion containing a point $t > cx_0^2$ has $x_1 \ge t > cx_0^2$. □

Lemma 8.3 (Superlinear horizons)**.** *Let $\psi : [1,\infty) \to (0,\infty)$ satisfy $\psi(x) \to \infty$. Then*

$$E_{\mathcal{F}_{x\psi(x)}}(X) = \#\{\nu : x_{0,\nu} \le X,\ x_{1,\nu} > x_{0,\nu}\psi(x_{0,\nu})\} = o(\log X).$$

Proof. Given $K > 0$ choose x_K with $\psi(x) \ge e^K$ for $x \ge x_K$. At most $x_K + 1$ members start below x_K. For consecutive members $a_j < a_{j+1}$ starting at or above x_K, $a_{j+1} \ge x_{1,j} > a_j\psi(a_j) \ge e^K a_j$, so $\log a_j$ increases by more than K at each step and there are at most $1 + \log X/K$ of them. Hence $\limsup E(X)/\log X \le 1/K$ for every K. □

Lemma 8.4 (Linear lacunarity)**.** *For every $\lambda > 1$ and $X \ge 2$,*

$$E_\lambda(X) = \#\{\nu : x_{0,\nu} \le X,\ x_{1,\nu} > \lambda x_{0,\nu}\} \le 2 + \frac{\log(X/2)}{\log\lambda}. \tag{8.4}$$

Proof. For starts $a_j \ge 2$, $a_{j+1} \ge x_{1,j} > \lambda a_j$, hence $a_j > 2\lambda^{j-1}$ for $j \ge 2$; add 1 for $(0,2]$. □

All four bounds are sharp up to constants for general families of disjoint intervals, as geometric and doubly geometric chains of intervals show.

8.2. **Arithmetic input: Kaczorowski–Pintz.** We separate the published statement, its translation from $M_0 = M + 2$ to M, and the consequences for excursions.

Theorem 8.5 (Kaczorowski–Pintz [9])**.** *Let $M_0(x) = M(x) - 1/\zeta(0) = M(x) + 2$, the function M_a of* [9] *at $a = 0$. Let $V(f,Y)$ be the number of sign changes of f on $[1,Y]$ counted through alternating-sign points, as defined in* [9]*. Then:*

(i) [9, Corollary 4, p. 178, (2.13)–(2.15)]

$$\liminf_{Y\to\infty} \frac{V(M_0,Y)}{\log Y} \ge \frac{\gamma_1}{\pi}, \tag{8.5}$$

with combined oscillation of size $x^{\Theta-\eta}$ for every $\eta > 0$, in the sense of their definition following (1.1): the alternating points t_j counted by V can be chosen with $|M_0(t_j)| \ge t_j^{\Theta-\eta}$;

(ii) *Corollary 4 is deduced from their Corollary 1, which asserts relations (2.1)–(2.3) of their Theorem 1* [9, pp. 175–176]*; in particular, for every $\delta > 0$ and $Y > Y_0(\delta)$, the interval $[Y^{1-\delta}, Y]$ contains two points t_+, t_- with $M_0(t_+) \ge t_+^{\Theta-\eta}$ and $M_0(t_-) \le -t_-^{\Theta-\eta}$.*

Lemma 8.6 (From M_0 to M)**.** *If $|M_0(t)| > 2$, then $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ and $|M(t)| \ge |M_0(t)| - 2$. In particular, for points with $|M_0(t)| \ge t^{\Theta-\eta}$ and t large, $\operatorname{sgn} M(t) = \operatorname{sgn} M_0(t)$ and $|M(t)| \ge \frac{1}{2}t^{\Theta-\eta}$.*

Proof. $M_0 > 2$ gives $M > 0$; $M_0 < -2$ gives $M < -4$. The second assertion uses $t^{\Theta-\eta} \to \infty$ for $\eta < \Theta$. □

Corollary 8.7 (Distinct large excursions)**.** *If $\Theta > 1/2$ and $0 < \eta < \Theta - 1/2$, then*

$$N_\eta(Y) \geq \Big(\frac{\gamma_1}{\pi} - o(1)\Big) \log Y, \tag{8.6}$$

where $N_\eta(Y)$ counts distinct excursions starting at most Y that contain a point $t \leq Y$ with $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$.

Proof. Take the alternating points of Theorem 8.5(i) in $[1, Y]$, discarding the boundedly many below the threshold of Lemma 8.6. By that lemma, consecutive points $t < t'$ have $M(t)$ and $M(t')$ of opposite signs. Since M is integer valued with steps $0, \pm 1$, it vanishes at some integer in (t, t'), so t and t' lie in different excursions. Each point with $M(t) \neq 0$ belongs to exactly one excursion, and excursions are ordered, so the excursions containing the ordered points are pairwise distinct. Their starts are below the points, hence at most Y, and $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$ at each point. □

Corollary 8.8 (Multiplicative localization of Mertens returns)**.** *For every fixed $A > 1$ there exists U_A such that every excursion starting at $u \geq U_A$ ends at $v \leq u^A$. Equivalently, M has only finitely many A-exceptional excursions; in particular only finitely many square-exceptional ones.*

Proof. Put $\delta = 1 - 1/A$ and $Y = u^A$, so that $[Y^{1-\delta}, Y] = [u, u^A]$, and fix $0 < \eta < \Theta$ (possible since $\Theta \geq 1/2$). For u large, Theorem 8.5(ii) and Lemma 8.6 give $t_+, t_- \in [u, u^A]$ with $M(t_+) > 0 > M(t_-)$. Neither point lies in $[u, u+1)$, because $M(u) = 0$; so $u < \lfloor t_\pm \rfloor$. Since M changes by $0, \pm 1$ at each integer, it vanishes at some integer n strictly between $\lfloor t_+ \rfloor$ and $\lfloor t_- \rfloor$, and $u < n < u^A$. If $u = x_{0,\nu}$, the first return satisfies $x_{1,\nu} \leq n < u^A$. □

Remark 8.9. A bare sign change of $M_0 = M + 2$ would not force a zero of M. The decisive point is that the local sign change is combined with oscillation of size $t^{\Theta-\eta} \to \infty$, so that Lemma 8.6 applies to both witnesses.

Remark 8.10 (Geometric counts and arithmetic localization). For the actual Mertens excursions Corollary 8.8 is stronger than Lemma 8.1: each A-exceptional family is finite. The geometric lemmas remain separate results because they are elementary, independent of ζ, and valid for every ordered family of disjoint intervals. This is exactly what the localization theorem below and the extreme-value theorems of Section 8.6 use: the former needs only a count below $(\gamma_1/\pi) \log X$, the latter needs a count valid pathwise in a model where no analogue of Kaczorowski–Pintz is available.

8.3. **Horizon localization.**

Theorem 8.11 (Horizon-localized RH criterion)**.** *Let h be a function on $[0, \infty)$ with $h(x) \geq x$, let $\mathcal{F}$ be a family of excursions, and suppose that for some $\kappa < \gamma_1/\pi$*

$$E_{\mathcal{F} \cup \mathcal{F}_h}(X) \leq (\kappa + o(1)) \log X \qquad (X \to \infty). \tag{8.7}$$

Then RH is equivalent to the following statement: for every $\varepsilon > 0$ there is C_ε such that

$$|M(t)| \leq C_\varepsilon t^{1/2+\varepsilon} \qquad \text{for all } t \in I_\nu,\ \nu \notin \mathcal{F},\ t \leq h(x_{0,\nu}). \tag{8.8}$$

No condition is imposed on the excursions in $\mathcal{F}$, nor on the part of any excursion lying beyond its horizon $h(x_0)$.

Proof. RH implies (8.8) by Littlewood's criterion. Conversely, suppose $\Theta > 1/2$ and fix $\eta, \varepsilon > 0$ with $1/2 + \varepsilon < \Theta - \eta$. By (8.6) there are at least $(\gamma_1/\pi - o(1)) \log X$ distinct excursions starting at most X, each containing a witness t with $|M(t)| \geq \frac{1}{2} t^{\Theta-\eta}$. Consider such an excursion ν with witness t. If $\nu \notin \mathcal{F}$ and $t \leq h(x_{0,\nu})$, the witness lies in the range of (8.8). Otherwise $\nu \in \mathcal{F}$, or $t > h(x_{0,\nu})$, in which case $x_{1,\nu} \geq t > h(x_{0,\nu})$ and $\nu \in \mathcal{F}_h$. By (8.7) at least

$$\Big(\frac{\gamma_1}{\pi} - \kappa - o(1)\Big) \log X \longrightarrow \infty$$

witnesses lie in the range of (8.8). They belong to distinct excursions, so they are unbounded, and at them $t^{\Theta-\eta}/t^{1/2+\varepsilon} \to \infty$, contradicting (8.8). □

The case $h(x) = \infty$ excludes a sparse family $\mathcal{F}$ of whole excursions; the case $\mathcal{F} = \emptyset$ truncates every excursion at its horizon.

Corollary 8.12 (Square horizon)**.** *For every fixed $A > 1$, RH is equivalent to* (8.8) *restricted to A-ordinary excursions. In particular*

$$\boxed{\text{RH} \iff |M(t)| \ll_\varepsilon t^{1/2+\varepsilon} \text{ for all } t \text{ in excursions with } x_1 \le x_0^2,} \tag{8.9}$$

and, equivalently, RH holds if and only if $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ for $x_{0,\nu} < t \le \min\{x_{1,\nu}, x_{0,\nu}^2\}$ over all excursions.

Proof. Take $\mathcal{F} = \mathcal{F}_{x^A}$, $h = \infty$, respectively $\mathcal{F} = \emptyset$, $h(x) = x^2$, in Theorem 8.11; Lemma 8.1 gives $\kappa = 0$. □

The count $E_\square(X) = O(\log\log X)$ is thus what makes the square horizon admissible. The square horizon can be lowered.

Corollary 8.13 (Descending horizons)**.** *RH is equivalent to Littlewood's bound $|M(t)| \ll_\varepsilon t^{1/2+\varepsilon}$ on the ranges*

$$x_{0,\nu} < t \le \min\{x_{1,\nu}, h(x_{0,\nu})\}$$

over all excursions, for each of the following horizons:

(i) *$h(x) = cx^2$, for any fixed $c > 0$;*
(ii) *$h(x) = x\psi(x)$, for any ψ with $\psi(x) \to \infty$;*
(iii) *$h(x) = \lambda x$, for any fixed $\lambda > \lambda_* := e^{\pi/\gamma_1} = 1.24889\ldots$*

The same holds with "truncate at $h(x_0)$" replaced by "discard the excursions with $x_1 > h(x_0)$". In particular it is enough to control the excursions with $x_1 \le \frac{5}{4}x_0$.

Proof. Apply Theorem 8.11 with $\mathcal{F} = \emptyset$ (respectively $\mathcal{F} = \mathcal{F}_h$, $h = \infty$). For (i), $c \ge 1$ reduces to Lemma 8.1 and $0 < c < 1$ to Lemma 8.2; $\kappa = 0$. For (ii), Lemma 8.3 gives $\kappa = 0$. For (iii), Lemma 8.4 gives $\kappa = 1/\log\lambda$, and $1/\log\lambda < \gamma_1/\pi$ exactly when $\lambda > \lambda_*$. Finally $5/4 > \lambda_*$. □

Remark 8.14 (Dyadic erosion and the uniform transition). For every fixed integer $j \ge 0$, Lemma 8.2 with $c = 2^{-j}$ gives

$$\#\left\{\nu : x_{0,\nu} \le X,\ x_{1,\nu} > \frac{x_{0,\nu}^2}{2^j}\right\} = O_{2^j}(\log\log X),$$

so Theorem 8.11 successively permits the horizons

$$x_0^2, \quad \frac{x_0^2}{2}, \quad \frac{x_0^2}{4}, \quad \frac{x_0^2}{8}, \ldots.$$

The constant in the fixed-c estimate depends on c; hence one may not set $j = j(x_0) \sim \log_2 x_0$ by that lemma alone. If instead $2^{j(x)} = x/\psi(x)$ with $\psi(x) \to \infty$, then the corresponding horizon is $x\psi(x)$ and Lemma 8.3 supplies the uniform $o(\log X)$ crossing count. Thus the rigorous erosion chain is

$$x_0^2 \to \frac{x_0^2}{2} \to \frac{x_0^2}{4} \to \cdots \to x_0\psi(x_0) \to \lambda x_0, \qquad \lambda > \lambda_*.$$

8.4. **Fixed-mask Catalan compression inside fractional square horizons.** The fractional-square localization has a useful interaction with the exact combinatorics of a completed excursion. The relevant Catalan time is not physical time: zero Möbius increments must first be removed. This support-time distinction is essential below.

Definition 8.15 (Support-time path and fixed-mask Catalan law)**.** Let $I = (x_0, x_1]$ be a complete excursion, and list the nonzero increments in I as

$$x_0 < a_1 < a_2 < \cdots < a_L \le x_1, \qquad \mu(a_j) \ne 0.$$

Put

$$r_I(t) := \#\{j : a_j \le t\}, \qquad x_0 < t \le x_1.$$

If $\sigma \in \{\pm 1\}$ is the orientation of the excursion, define

$$S_0 = 0, \qquad S_r := \sigma \sum_{j=1}^{r} \mu(a_j) \qquad (1 \le r \le L).$$

Then $L = 2n$ is even and

$$S_0 = S_L = 0, \qquad S_r > 0 \quad (1 \le r < L), \qquad S_r - S_{r-1} \in \{-1, +1\}.$$

Thus $(S_r)_{0 \le r \le L}$ is a primitive Dyck path. The *fixed-mask primitive Catalan law* keeps the physical support $\{a_1, \dots, a_L\}$ fixed and assigns to those support positions a uniformly chosen primitive Dyck sign sequence of length L. We denote its probability by $\mathbb{P}_{\mathrm{pCat},I}$.

For the observed excursion one has, exactly,

$$|M(t)| = S_{r_I(t)}, \qquad r_I(t) \le t - x_0 < t. \tag{8.10}$$

The first equality uses the orientation of the excursion and the fact that M is constant across the zero increments.

We first record the global Catalan estimate, since it is useful in its own right and clarifies why a different estimate is needed at the square horizon.

Lemma 8.16 (Catalan maximum tail)**.** *Let $L = 2n$ and let $H_{\max}$ be the maximum of a uniformly chosen Dyck path of length $2n$. For $1 \le h \le n$,*

$$\mathbb{P}_{\mathrm{Cat},L}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le (n+1)\exp\left(-\frac{h^2}{n+h}\right) \le \left(\frac{L}{2}+1\right)e^{-h^2/L}. \tag{8.11}$$

For a uniformly chosen primitive Dyck path of length $L = 2n$,

$$\mathbb{P}_{\mathrm{pCat},L}\{H_{\max} \ge h\} \le \frac{L}{2}\exp\left(-\frac{(h-1)^2}{L-2}\right), \qquad 2 \le h \le n. \tag{8.12}$$

Proof. There are $C_n = (n+1)^{-1}\binom{2n}{n}$ Dyck paths of length $2n$ and $\binom{2n}{n}$ unrestricted bridges. Reflecting a bridge after its first visit to level h gives a bijection between bridges that reach h and unrestricted walks ending at $2h$, hence their number is $\binom{2n}{n+h}$. Therefore

$$\mathbb{P}_{\mathrm{Cat},2n}\{H_{\max} \ge h\} \le (n+1)\frac{\binom{2n}{n+h}}{\binom{2n}{n}}.$$

Moreover

$$\frac{\binom{2n}{n+h}}{\binom{2n}{n}} = \prod_{j=1}^{h}\frac{n-j+1}{n+j},$$

and $\log(1-x) \le -x$ gives

$$\log\frac{\binom{2n}{n+h}}{\binom{2n}{n}} \le -\sum_{j=1}^{h}\frac{2j-1}{n+j} \le -\frac{h^2}{n+h}.$$

Since $n+h \le 2n = L$, (8.11) follows. A primitive path of length $2n$ has the form $+1$, followed by a Dyck path of length $2n-2$, followed by -1; its maximum is one plus the maximum of the interior Dyck path. Applying the preceding bound with $n-1$ and $h-1$ proves (8.12). □

Remark 8.17 (Why the maximum tail is not the square-horizon estimate). The exponent in (8.12) is of order H^2/L. For a general family of disjoint intervals retained by the square horizon one may have $L \asymp x_0^2$ (for M itself such excursions are finitely many by Corollary 8.8), while the relevant witness lies much earlier, at physical time $t \asymp x_0$. Then a violation of size $t^{1/2+\varepsilon}$ need not make H^2/L large. Thus the global maximum tail, although correct, is not the right Catalan quantity for combining with the fractional-square horizon. The correct quantity is the pointwise height at the support time $r_I(t)$.

Lemma 8.18 (Pointwise primitive-Catalan tail). *Let $L = 2n \geq 4$ and let (S_r) be a uniformly chosen primitive Dyck path of length L. For $1 \leq r \leq L-1$ and $h \geq 1$,*

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \geq h\} \leq 8n^{3/2} \exp\left(-\frac{h^2}{2r}\right) \leq 3L^{3/2} \exp\left(-\frac{h^2}{2r}\right). \tag{8.13}$$

By time reversal the same estimate holds with r replaced by $L-r$; hence

$$\mathbb{P}_{\mathrm{pCat},L}\{S_r \geq h\} \leq 3L^{3/2} \exp\left[-\frac{h^2}{2\min\{r, L-r\}}\right]. \tag{8.14}$$

If $h > \min\{r, L-r\}$ the probability is 0, so the bound is only relevant for $h \leq \min\{r, L-r\}$.

Proof. Ignore both positivity and the endpoint constraint. Among all 2^L sign words, the proportion whose first r signs have sum at least h is at most $\exp(-h^2/(2r))$ by the elementary Chernoff bound for a Rademacher sum. Therefore the number of primitive Dyck paths satisfying $S_r \geq h$ is at most

$$2^L \exp\left(-\frac{h^2}{2r}\right).$$

The total number of primitive Dyck paths of length $2n$ is C_{n-1}. The standard central-binomial lower bound $\binom{2m}{m} \geq 4^m/(2\sqrt{m})$ gives

$$C_{n-1} = \frac{1}{n}\binom{2n-2}{n-1} \geq \frac{4^{n-1}}{2n\sqrt{n-1}}.$$

Since $2^L = 4^n$, division gives the first inequality in (8.13); the second follows from $8n^{3/2} < 3(2n)^{3/2}$. Reversing the path and changing the signs of the reversed increments preserves the primitive Dyck class and carries height S_r to height S_{L-r}, which proves (8.14). □

For an observed excursion I define its pointwise Catalan upper-tail value by

$$p_{\mathrm{Cat}}(I,t) := \mathbb{P}_{\mathrm{pCat},I}\left\{S_{r_I(t)} \geq |M(t)|\right\}. \tag{8.15}$$

This is a tail probability under the fixed-mask Catalan reference law; it is not a probability assigned to the arithmetic Möbius path itself.

Corollary 8.19 (Fractional-square Catalan rarity). *Fix $c > 0$ and $\varepsilon > 0$. Let $I = (x_0, x_1]$ be a complete excursion with $x_1 \leq cx_0^2$, and keep its zero mask fixed. For all sufficiently large x_0,*

$$\mathbb{P}_{\mathrm{pCat},I}\left\{\exists\, t \in I :\ S_{r_I(t)} \geq t^{1/2+\varepsilon}\right\} \leq 3c^{5/2}x_0^5 \exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right). \tag{8.16}$$

In particular the right-hand side is smaller than x_0^{-A} for every fixed $A > 0$ once x_0 is large enough.

Proof. At a support position $t = a_r$ one has $r \leq t - x_0 < t$. Hence Lemma 8.18 with $h = t^{1/2+\varepsilon}$ gives

$$\mathbb{P}_{\mathrm{pCat},I}\{S_r \geq t^{1/2+\varepsilon}\} \leq 3L^{3/2} \exp\left(-\frac{1}{2}t^{2\varepsilon}\right) \leq 3L^{3/2} \exp\left(-\frac{1}{2}x_0^{2\varepsilon}\right).$$

There are at most L support positions. Since $L \leq x_1 - x_0 \leq x_1 \leq cx_0^2$, the union bound gives (8.16). Between support positions the path is constant, so no additional physical times have to be counted. □

The preceding statement is purely combinatorial: it measures how far a large value lies in the tail of a reference law, not how the Möbius constraints select a path. Combined with the localization theorem it gives the following quantitative form.

Corollary 8.20 (Catalan surprise of a violation)**.** *Suppose $\Theta > 1/2$, fix $0 < \eta < \Theta - \frac{1}{2}$, put $\delta = \Theta - \eta - \frac{1}{2}$, and fix $\lambda > \lambda_*$. Then there are infinitely many excursions $I = (x_0, x_1]$ with $x_1 \le \lambda x_0$ containing a point t with*

$$-\log p_{\mathrm{Cat}}(I,t) \;\ge\; \frac{1}{8}x_0^{2\delta} - O(\log x_0). \tag{8.17}$$

The same holds with $x_1 \le \lambda x_0$ replaced by $x_1 \le c x_0^2$ for any fixed $c > 0$.

Proof. By Corollary 8.7 and Lemma 8.4, at least $(\gamma_1/\pi - 1/\log\lambda - o(1))\log X$ excursions starting at most X have $x_1 \le \lambda x_0$ and contain t with $h = |M(t)| \ge \frac{1}{2}t^{1/2+\delta}$; for cx_0^2 use Lemma 8.2 instead. With $r = r_I(t) < t$ and $L \le x_1 - x_0$, Lemma 8.18 gives $p_{\mathrm{Cat}}(I,t) \le 3L^{3/2}\exp(-h^2/(2r)) \le 3L^{3/2}\exp(-t^{2\delta}/8)$, and $t > x_0$, $L \le \lambda x_0$ (respectively $L \le cx_0^2$). □

Remark 8.21 (Selector bounds are pointwise bounds). Up to polynomial factors, $p_{\mathrm{Cat}}(I,t)$ is a decreasing function of $|M(t)|^2/r_I(t)$: by Lemma 8.18,

$$-\log p_{\mathrm{Cat}}(I,t) \;\ge\; \frac{|M(t)|^2}{2r_I(t)} - \log\bigl(3L^{3/2}\bigr).$$

Hence a selector bound $-\log p_{\mathrm{Cat}}(I,t) \le \Phi(x_0)$ with $\Phi(x) = x^{o(1)}$ on the excursions with $x_1 \le \lambda x_0$ implies $|M(t)| \le t^{1/2}x_0^{o(1)}$ there, and RH then follows from Corollary 8.13; it is not an independent criterion. The selector bound is strictly stronger than Littlewood's bound on these ranges: it controls $|M(t)|$ at the diffusive scale $r_I(t)^{1/2}$ of the support time, which may be much smaller than $t^{1/2}$, and a polynomial selector $p_{\mathrm{Cat}} \ge x_0^{-A}$ would give $|M(t)| \ll \sqrt{t\log t}$ on the retained ranges, beyond any bound known to follow from RH. Turning p_{Cat} into an exact measure of deviation would require the matching lower bound $-\log p_{\mathrm{Cat}}(I,t) \ll |M(t)|^2/r_I(t) + \log L$ for the primitive Catalan tail, which is not proved here.

Remark 8.22 (The correct quantifier). A single value $|M(t)| \ge t^{1/2+\varepsilon}$, or many values with bounded ratio $|M(t)|/t^{1/2+\varepsilon}$, is compatible with RH, since Littlewood's criterion allows a constant C_ε. The negation of the criterion is

$$\exists\,\varepsilon_0 > 0: \qquad \sup_{t\ge 1}\frac{|M(t)|}{t^{1/2+\varepsilon_0}} = \infty,$$

equivalently there are $t_j \to \infty$ and $A_j \to \infty$ with $|M(t_j)| \ge A_j t_j^{1/2+\varepsilon_0}$. Accordingly, the rarity bound (8.16) is a statement about the reference law, and the arithmetic content lies in the sequence form below. If $\Theta > 1/2$ and $0 < \delta < \Theta - \frac{1}{2}$, the oscillation theorem gives $|M(t_j)| \ge \frac{1}{2}t_j^{1/2+\delta}$ along a sequence, and then $|M(t_j)|/t_j^{1/2+\varepsilon} \ge \frac{1}{2}t_j^{\delta-\varepsilon} \to \infty$ for every $0 < \varepsilon < \delta$.

Corollary 8.23 (Catalan cost along a Littlewood-violating sequence)**.** *Let $t_j \to \infty$, $A_j \to \infty$ and $\varepsilon_0 > 0$ with $|M(t_j)| \ge A_j t_j^{1/2+\varepsilon_0}$, and let $I_j = (x_{0,j}, x_{1,j}]$ be the excursion containing t_j. If $x_{1,j} \le x_{0,j}^B$ for a fixed B, then*

$$-\log p_{\mathrm{Cat}}(I_j,t_j) \;\ge\; \frac{A_j^2}{2}\,t_j^{2\varepsilon_0} - O_B(\log x_{0,j}) \;\ge\; \frac{A_j^2}{2}\,x_{0,j}^{2\varepsilon_0} - O_B(\log x_{0,j}).$$

Proof. With $r = r_{I_j}(t_j) < t_j$ and $h = |M(t_j)|$, $h^2/(2r) \ge A_j^2 t_j^{1+2\varepsilon_0}/(2t_j)$; apply Lemma 8.18 with $L \le x_{1,j} \le x_{0,j}^B$. □

Selection energies: complete excursions and open prefixes. The support-time clock of Definition 8.15 is the squarefree-compressed clock

$$Q(x) := \#\{n \le x : \mu(n) \neq 0\}, \qquad L^{\#}(I) = Q(x_1) - Q(x_0) \le x_1 - x_0,$$

and $r_I(t) = Q(t) - Q(x_0)$. Two tail quantities are natural: one for the height of a complete excursion, and one for an open prefix, which does not require knowledge of the future return.

Definition 8.24 (Selection energies)**.** For a complete excursion I with height $H(I)$ and compressed length $L^{\#} = L^{\#}(I)$, put

$$p_{\mathrm{Cat}}^{\max}(I) := \mathbb{P}_{\mathrm{pCat},L^{\#}}\{H_{\max} \ge H(I)\}, \qquad \mathcal{I}_{\mathrm{Cat}}(I) := -\log p_{\mathrm{Cat}}^{\max}(I).$$

For $t \ge 1$ with $M(t) \neq 0$ let $a(t)$ be the last zero of M before t and $r = Q(t) - Q(a(t))$. After orientation, the compressed segment from $a(t)$ to t is a positive meander of length r: a ± 1 path $(S_k)_{k \le r}$ with $S_0 = 0$ and $S_k > 0$ for $1 \le k \le r$. Let $\mathcal{M}_r$ be the set of such paths and

$$p_{\mathrm{open}}(t) := \mathbb{P}_{\mathcal{M}_r}\{S_r \ge |M(t)|\},$$

the uniform measure on $\mathcal{M}_r$.

Lemma 8.25 (Height–entropy inequalities)**.** (i) *For every complete excursion with $L^{\#} \ge 4$,*

$$\mathcal{I}_{\mathrm{Cat}}(I) \;\ge\; \frac{(H(I)-1)^2}{L^{\#}-2} - \log\frac{L^{\#}}{2}.$$

(ii) $\#\mathcal{M}_r = \binom{r-1}{\lfloor (r-1)/2 \rfloor} \ge 2^r/(4\sqrt{r})$*, and for $h \ge 1$*

$$\mathbb{P}_{\mathcal{M}_r}\{S_r \ge h\} \le 4\sqrt{r}\, e^{-h^2/(2r)}, \qquad -\log p_{\mathrm{open}}(t) \;\ge\; \frac{M(t)^2}{2r} - \log\bigl(4\sqrt{r}\bigr).$$

Proof. (i) is (8.12). (ii) The count of positive meanders is classical (ballot theorem; the number ending at s is $\frac{s}{r}\binom{r}{(r+s)/2}$). From $\binom{2m}{m} \ge 4^m/(2\sqrt{m})$ and $\binom{2m+1}{m} = \frac{1}{2}\binom{2m+2}{m+1}$ one gets $\#\mathcal{M}_r \ge 2^r/(4\sqrt{r})$. Among all 2^r words the proportion with $S_r \ge h$ is at most $e^{-h^2/(2r)}$ by the Chernoff bound; divide. □

Proposition 8.26 (Selector criteria)**.** (i) Complete excursions. *Fix $\lambda > \lambda_*$. If there is A such that $p_{\mathrm{Cat}}^{\max}(I) \ge x_0^{-A}$ for every sufficiently large complete excursion with $x_1 \le \lambda x_0$, then $H(I) \ll_{A,\lambda} \sqrt{x_0 \log x_0}$ on that family, and RH holds.*

(ii) Open prefixes. *If there is A such that $p_{\mathrm{open}}(t) \ge t^{-A}$ for all large t with $M(t) \neq 0$, then $|M(t)| \ll_A \sqrt{t \log t}$, and RH holds.*

Proof. (i) By Lemma 8.25(i), $(H-1)^2 \le (L^{\#}-2)(A\log x_0 + \log L^{\#})$, and $L^{\#} \le (\lambda - 1)x_0$. Littlewood's bound therefore holds on every excursion with $x_1 \le \lambda x_0$, and Corollary 8.13(iii), in the form that discards the λ-long excursions, gives RH. (ii) By Lemma 8.25(ii), $M(t)^2 \le 2r(A\log t + \log(4\sqrt{r}))$ with $r \le t$. □

Remark 8.27 (Status of the two selectors). Both hypotheses are stronger than RH: each yields $|M| \ll \sqrt{t\log t}$ on its range, beyond any bound known to follow from RH. Neither follows from RH alone: a ballistic stretch of compressed length r has $-\log p \asymp r$, and RH does not exclude such stretches with r a small power of t. The open selector controls every point without reference to the future return; the complete selector uses only the horizon-localized family, where $L^{\#} \le (\lambda-1)x_0$ is controlled. By contraposition and Corollary 8.23, failure of RH forces the compressed Möbius path to realize, along an unbounded sequence, sectors of Catalan or meander weight $\exp(-c\,t^{2\delta})$ with $\delta > 0$, not merely polynomially rare ones.

Case (iii) implies (i) and (ii): in both, $h(x_0) \geq \lambda x_0$ for large x_0, so the bound on the larger ranges implies it on the ranges of (iii), the finitely many remaining excursions affecting only C_ε. The horizon has been lowered from x_0^2 to λx_0: the possible failure of Littlewood's bound, if any, is visible within a bounded multiplicative distance of the start of an excursion. Quantitatively:

Corollary 8.28 (Two-sided localization of large values)**.** *Suppose $\Theta > 1/2$, and fix $0 < \eta < \Theta - \frac{1}{2}$ and $\lambda > \lambda_*$. Among the excursions starting at $x_0 \leq X$, at least*

$$\Big(\frac{\gamma_1}{\pi} - \frac{1}{\log \lambda} - o(1)\Big) \log X$$

contain a point t_* with $|M(t_*)| \geq \frac{1}{2} t_*^{\Theta-\eta}$ and

$$x_0 + \tfrac{1}{2} x_0^{\Theta-\eta} \leq t_* \leq \lambda x_0.$$

Proof. Of the excursions of Corollary 8.7, those whose witness exceeds λx_0 are λ-long, hence at most $\log X / \log \lambda + 2$ by Lemma 8.4. For the others, $M(x_0) = 0$ and unit steps give $|M(t_*)| \leq t_* - x_0$, so $t_* - x_0 \geq \frac{1}{2} t_*^{\Theta-\eta} \geq \frac{1}{2} x_0^{\Theta-\eta}$. □

Remark 8.29 (The threshold λ_*). The function $\cos(\gamma_1 \log x)$ changes sign γ_1/π times per unit of $\log x$, i.e. once in every multiplicative interval of ratio $e^{\pi/\gamma_1} = \lambda_*$; the constant in (8.5) is this density. On the geometric side, a chain of λ-long intervals realizes the bound of Lemma 8.4. The method of Theorem 8.11 compares these two counts, and λ_* is the exact point where they balance: for $\lambda \leq \lambda_*$ the λ-long family can, as far as counting alone is concerned, absorb every Kaczorowski–Pintz excursion. Going below λ_* requires information on where inside an excursion the large values occur, not only how many excursions carry them.

8.5. **The linear window and the one-step recursion.** Once the horizon is below $2x_0$, the square-range bilinear structure of Proposition 8.65 is no longer needed: on $(u, 2u]$ the Möbius function is a *linear* expression in the prefix $\mu \mathbf{1}_{[1,u]}$.

Proposition 8.30 (Linear window)**.** *Let $u \geq 1$. For every integer $u < n \leq 2u$,*

$$\mu(n) = -\sum_{\substack{d \mid n \\ d \leq u}} \mu(d). \tag{8.18}$$

Consequently, if $M(u) = 0$ and $u < t \leq 2u$,

$$M(t) = -\sum_{d \leq u} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{8.19}$$

Proof. Since $n \geq 2$, $\sum_{d \mid n} \mu(d) = 0$. Every proper divisor of n is at most $n/2 \leq u$, and every divisor $d \leq u$ is proper because $n > u$. Summing (8.18) over $u < n \leq t$, the number of multiples of d in $(u, t]$ is $\lfloor t/d \rfloor - \lfloor u/d \rfloor$. □

Corollary 8.31 (Localized prefix formula)**.** *Let $M(u) = 0$, let $1 < \lambda < 2$, and let $u < t \leq \lambda u$. Then*

$$M(t) = -\sum_{d \leq \lambda u/2} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right). \tag{8.20}$$

Indeed, if $\lambda u/2 < d \leq u$, then $d > u/2$ and the first multiple of d strictly larger than u is $2d > \lambda u \geq t$.

Corollary 8.32 (Prefix-dependency cutoff)**.** *Put*

$$\theta_* := \frac{1}{2} e^{\pi/\gamma_1} = 0.6244\ldots.$$

For every fixed θ with $\theta_* < \theta < 1$, RH is equivalent to Littlewood's bound on every excursion starting at u throughout

$$u < t \le \min\{x_1, 2\theta u\},$$

where, throughout that window, $M(t)$ is given by the sum (8.20) using only the prefix $\mu(d)$ with $d \le \theta u$. In particular, the choice $\theta = 5/8$ corresponds to $\lambda = 5/4$ and gives the RH-equivalent window

$$1 \le r = t - u \le \min\{\ell - 1, u/4\}$$

using only $d \le 5u/8$.

The number θ_* is an infimum, not an attained endpoint, because the horizon theorem requires $\lambda > \lambda_*$.

Proof. Take $\lambda = 2\theta > \lambda_*$ in Corollary 8.13(iii), then use Corollary 8.31. □

Corollary 8.33 (Algebraic dependency gap)**.** *Let $\theta_* < \theta < 1$. If u is a zero of M, then every increment $\mu(n)$ with*

$$u < n \le 2\theta u$$

is determined by the prefix $\mu(1), \dots, \mu(\lfloor \theta u \rfloor)$. Consequently the intermediate block

$$(\theta u, u]$$

does not occur as a direct input in the divisor recursion determining the localized future window. As $\theta \downarrow \theta_*$, the omitted block has relative length tending to $1 - \theta_* = 0.3755\ldots$.

Proof. For $u < n \le 2\theta u$, every proper divisor $d \mid n$ satisfies $d \le n/2 \le \theta u$. Hence

$$\mu(n) = -\sum_{\substack{d \mid n \\ d < n}} \mu(d)$$

uses only $d \le \theta u$. □

Remark 8.34 (Direct dependency is not independence). Corollary 8.33 is an algebraic statement about which values occur as inputs of the divisor recursion. It does not assert statistical or arithmetic independence between the omitted block $(\theta u, u]$ and the earlier prefix.

In convolution form, (8.18) says that $(\mu - \mu_u) * 1 = \mu \mathbf{1}_{(u,\infty)}$ on $n \le 2u$, the order-one analogue of (8.30), which for $u < n \le u^2$ reads $\mu(n) = -(\mu_u * \mu_u * 1)(n)$; on $(u, 2u]$ both give $\mu(n)$, the bilinear form being the quadratic continuation of the linear one. Equivalently, (8.19) follows from $\sum_{d \le t} \mu(d) \lfloor t/d \rfloor = 1$, since $\lfloor t/d \rfloor = 1$ for $u < d \le t \le 2u$.

Proposition 8.35 (One-step recursion)**.** *For real $x \ge 1$,*

$$\sum_{k \le x} M\Big(\frac{x}{k}\Big) = 1. \tag{8.21}$$

Hence, if $M(u) = 0$ and $t > u$,

$$M(t) = -\sum_{2 \le k \le t} \left[M\left(\frac{t}{k}\right) - M\Big(\frac{u}{k}\Big) \right], \tag{8.22}$$

with $M(y) = 0$ for $y < 1$. Every argument on the right is at most $t/2$; hence on a localized window $u < t \le \lambda u$ all arguments are at most $\lambda u/2$. For $\lambda = 5/4$ this is $5u/8$.

Proof. $\sum_{k \le x} M(x/k) = \sum_{k \le x} \sum_{d \le x/k} \mu(d) = \sum_{m \le x} \sum_{d \mid m} \mu(d) = 1$. Subtract the identity at u, where $M(u) = 0$, and note $M(u/k) = 0$ for $k > u$. □

Proposition 8.36 (Scale cascade in the localized excursion window)**.** *Let $(u, v]$ be an excursion, let $\lambda_* < \lambda < 2$, and let $u < t \le \min\{v, \lambda u\}$. For $k \ge 2$ put*

$$\Delta_k(u,t) := M(t/k) - M(u/k),$$

and for an integer $K \ge 2$ put

$$R_K(u,t) := -\sum_{K<k\le t} \Delta_k(u,t).$$

Then, exactly,

$$M(t) = -\sum_{k=2}^{K} \Delta_k(u,t) + R_K(u,t), \tag{8.23}$$

and $R_K(u,t)$ uses only values $M(q)$ with

$$q \le \frac{\lambda u}{K+1}.$$

Equivalently, all dependence on values above $\lambda u/(K+1)$ is concentrated in the $K-1$ increments $\Delta_2, \dots, \Delta_K$. Moreover

$$\Delta_k(u,t) = \sum_{u/k<n\le t/k} \mu(n),$$

so the supporting interval has length $(t-u)/k$.

Proof. Equation (8.23) is Proposition 8.35 with the terms $2 \le k \le K$ separated. If $k > K$, then $t/k \le \lambda u/(K+1)$ and $u/k \le \lambda u/(K+1)$. The last identity is the definition of the summatory function on the interval $(u/k, t/k]$. □

Corollary 8.37 (First channels)**.** *For $K = 2$,*

$$M(t) = -\big(M(t/2) - M(u/2)\big) + R_2(u,t),$$

where R_2 uses only $q \le \lambda u/3$. With the convenient localized choice $\lambda = 5/4$, the only channel above $5u/12$ is

$$\Delta_2(u, u+r) = \sum_{u/2<n\le u/2+r/2} \mu(n), \qquad 0 < r \le u/4,$$

whose supporting interval has length at most $u/8$. In the limiting sense $\lambda \downarrow \lambda_*$, the successive remainder cutoffs are

$$\frac{\lambda_*}{2}u = 0.6244\ldots u, \quad \frac{\lambda_*}{3}u = 0.4162\ldots u, \quad \frac{\lambda_*}{4}u = 0.3122\ldots u, \quad \frac{\lambda_*}{5}u = 0.2497\ldots u.$$

Lemma 8.38 (Channel persistence extracted from a high excursion)**.** *Let $S_r = \sigma M(u+r) > 0$ on $1 \le r \le h$ with $S_0 = 0$ and $|S_r - S_{r-1}| \le 1$. Fix $K \ge 2$ and write, using Proposition 8.36,*

$$S_r = \sum_{k=2}^{K} X_k(r) + Y_K(r), \quad X_k(r) := -\sigma\Delta_k(u, u+r), \quad Y_K(r) := \sigma R_K(u, u+r).$$

If $H = \max_{r\le h} S_r$ and τ is a point with $S_\tau = H$, then among the last $\lfloor H/2 \rfloor + 1$ indices

$$\tau - \lfloor H/2 \rfloor, \dots, \tau$$

there is one fixed component among $X_2, \dots, X_K, Y_K$ which is at least $H/(2K)$ at no fewer than $\lfloor H/(2K) \rfloor$ indices.

Proof. Since $S_0 = 0$ and the path has unit steps, $\tau \ge H$ and $S_{\tau-j} \ge H - j \ge H/2$ for $0 \le j \le \lfloor H/2 \rfloor$. At every one of these indices the sum of the K displayed components is at least $H/2$, hence some component is at least $H/(2K)$. Pigeonholing the indices among the K components gives the claim. □

Remark 8.39 (What the persistence lemma adds). Lemma 8.38 is not a bound for M. It converts a large value attained inside one excursion into a quantified persistence requirement: either one of the finitely many scaled interval sums $\Delta_2, \dots, \Delta_K$ repeatedly carries a positive fraction of the height, or the responsibility lies repeatedly in the deeper remainder, which is supported entirely below $\lambda u/(K+1)$. This avoids the iteration defect of Remark 8.51: no new point is claimed to be the start-window point of another excursion.

Corollary 8.40 (Channel descent with explicit loss). *Let u be a zero of M, $t > u$, $K \geq 2$, and $0 < \beta < 1$. Then either*

(i) *there are $2 \leq k \leq K$ and $q \in \{\lfloor u/k \rfloor, \lfloor t/k \rfloor\}$ with $|M(q)| \geq |M(t)|/(2K)$, hence*

$$\frac{|M(q)|}{q^\beta} \geq \frac{k^\beta}{2K} \cdot \frac{|M(t)|}{t^\beta}, \qquad q \leq \frac{t}{k};$$

or

(ii) *$|R_K(u,t)| \geq |M(t)|/K$, where R_K depends only on $M(q)$ with $q \leq t/(K+1)$.*

The statement holds at every $t > u$, not only in a localized window.

Proof. The proof of Proposition 8.36 applies verbatim to every $t > u$, with R_K supported on $q \leq t/(K+1)$. By (8.23), one of the K terms $\Delta_2, \dots, \Delta_K, R_K$ has modulus at least $|M(t)|/K$. If it is $\Delta_k = M(\lfloor t/k \rfloor) - M(\lfloor u/k \rfloor)$, one of its two values has modulus at least $|M(t)|/(2K)$, and $q \leq t/k$. □

Remark 8.41 (Channels do not iterate to a bound). Corollary 8.40 has no iteration gap: it applies at the new point q relative to the last zero of M before q. Its defect is the constant loss $k^\beta/(2K) < 1$ per step. If alternative (i) held at every step, a chain of length $m \asymp \log t$ would give only $|M(t)| \leq (2K)^m\, O(1) = t^{O(\log K)}$, weaker than the trivial bound. A closing argument must therefore show that alternative (ii), or alternative (i) with loss factor at least 1, occurs along the chain; both are statements about cancellation, not about dependency.

Remark 8.42 (What the recursion determines). Iterating (8.21) determines all of M from $M(1) = 1$, so determinacy alone carries no information; cf. Remark 9.2. The content of (8.22), after Corollary 8.32, is more precise: at the start u of an excursion, every value in an RH-equivalent window $u < t \leq \lambda u$ is a signed sum of increments of M over scaled intervals $(u/k, t/k]$ lying below $\lambda u/2$. This is a descent of dependency only; no descent of normalized size follows from the identity.

Corollary 8.43 (RH in the localized linear window). *RH holds if and only if, for every $\varepsilon > 0$,*

$$\left| \sum_{d \leq 5u/8} \mu(d) \left(\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{u}{d} \right\rfloor \right) \right| = \left| \sum_{2 \leq k \leq t} \left[M\left(\frac{t}{k}\right) - M\left(\frac{u}{k}\right) \right] \right| \ll_\varepsilon t^{1/2+\varepsilon}$$

for every excursion starting at u and every

$$u < t \leq \min\left\{ x_1, \frac{5}{4}u \right\}.$$

Equivalently, in intra-excursion coordinates $t = u + r$ it is enough to control

$$1 \leq r \leq \min\{\ell - 1, u/4\}.$$

Proof. Use Corollary 8.32 with $\theta = 5/8$, together with Proposition 8.35. □

Proposition 8.44 (Kernel form of the recursion). *Let $M(u) = 0$ and let t be an integer with $u < t < 2u$. For $1 \leq q \leq t/2$ put*

$$c_q(t,u) = \left(\left\lfloor \frac{t}{q} \right\rfloor - \left\lfloor \frac{t}{q+1} \right\rfloor \right) - \left(\left\lfloor \frac{u}{q} \right\rfloor - \left\lfloor \frac{u}{q+1} \right\rfloor \right).$$

Then:

(i) $M(t) = -\sum_{1\le q\le t/2} c_q(t,u)M(q)$;

(ii) $c_q(t,u) = \dfrac{t-u}{q(q+1)} + \rho_q$ *with* $|\rho_q| < 2$; *for* $q > \sqrt{t}$ *one has* $c_q \in \{-1,0,1\}$, *and* $c_q \neq 0$ *only if* $q = \lfloor t/k \rfloor$ *or* $q = \lfloor u/k \rfloor$ *for some* $k < \sqrt{t}$;

(iii) $c_1(t,u) \ge \frac{1}{2}(t-u-1)$;

(iv) $M(t) = -(t-u)\,\Theta(\sqrt{t}) + E(t,u)$, *where* Θ *is the coefficient* (10.10) *and*

$$|E(t,u)| \le 2\sum_{q\le\sqrt{t}} |M(q)| + \sum_{k<\sqrt{t}} \Big(\Big|M\Big(\frac{t}{k}\Big)\Big| + \Big|M\Big(\frac{u}{k}\Big)\Big|\Big).$$

Proof. (i) In (8.22) group the indices $k \ge 2$ by the value $q = \lfloor t/k \rfloor$, respectively $q = \lfloor u/k \rfloor$. The number of $k \ge 1$ with $\lfloor t/k \rfloor = q$ is $\lfloor t/q \rfloor - \lfloor t/(q+1) \rfloor$, and for $q \le t/2$ none of them is $k = 1$. The same holds for u, because $k = 1$ gives $q = u > t/2$. (ii) Each bracket equals $x/(q(q+1))$ minus a difference of two fractional parts, so it differs from $x/(q(q+1))$ by less than 1. For $q > \sqrt{t} \ge \sqrt{u}$ both $t/q - t/(q+1)$ and $u/q - u/(q+1)$ are < 1, so each bracket is 0 or 1, and it is 1 only if $q = \lfloor x/k \rfloor$ with $k \le x/q < \sqrt{t}$. (iii) $c_1 = (t-u) - (\lfloor t/2 \rfloor - \lfloor u/2 \rfloor)$ and $\lfloor t/2 \rfloor - \lfloor u/2 \rfloor \le \lceil (t-u)/2 \rceil$. (iv) Insert (ii) into (i) for $q \le \sqrt{t}$; the smooth part is $(t-u)\sum_{q\le\sqrt{t}} M(q)/(q(q+1)) = (t-u)\Theta(\sqrt{t})$. For $q > \sqrt{t}$ use the last assertion of (ii). □

Corollary 8.45 (Loss of the pointwise induction). *Let* $0 < \beta < 1$ *and suppose* $|M(q)| \le Cq^\beta$ *for all* $q \le t/2$. *Then, for* $u < t < 2u$ *with* $M(u) = 0$,

$$\big|M(t) + (t-u)\Theta(\sqrt{t})\big| \ll_\beta C\,t^{(1+\beta)/2}, \tag{8.24}$$

while the termwise bound of the full kernel gives only $|M(t)| \ll_\beta C\,t$. *For* $\beta = \frac{1}{2} + \varepsilon$ *the pointwise induction therefore loses* $t^{1/2-\varepsilon}$, *and* $t^{1/4-\varepsilon/2}$ *after the smooth component* $-(t-u)\Theta(\sqrt{t})$ *is extracted.*

Proof. By Proposition 8.44(iv), $|E| \le 2C\sum_{q\le\sqrt{t}} q^\beta + 2Ct^\beta \sum_{k<\sqrt{t}} k^{-\beta} \ll_\beta Ct^{(1+\beta)/2}$. The full kernel satisfies $\sum_q |c_q| q^\beta \ge c_1 \ge \frac{1}{2}(t-u-1)$ by (iii), and $\sum_q |c_q| q^\beta \ll_\beta t$ by (ii). □

The smooth component is not small for termwise reasons: by (10.12),

$$\Theta(\sqrt{t}) = -\sum_{n>\sqrt{t}} \frac{M(n)}{n(n+1)},$$

so its size is governed by the global identity $\sum_{n\ge1} M(n)/(n(n+1)) = 0$, i.e. by the pole of ζ at $s = 1$ (Remark 10.14), and involves values of M beyond $t/2$. Even under RH the tail gives only $(t-u)\Theta(\sqrt{t}) \ll t^{3/4+\varepsilon/2}$, the same order as the bound for $E(t,u)$; the RH scale for $M(t)$ is a statement about their sum.

Remark 8.46 (Localized intra-excursion system). Let $I = (u, x_1]$ be an excursion of length $\ell = x_1 - u$ and orientation σ, and put

$$S_r := \sigma M(u+r) > 0 \qquad (1 \le r < \ell), \qquad S_0 = S_\ell = 0.$$

For a fixed $\lambda \in (\lambda_*, 2)$ set

$$h_\lambda(u) := \min\{\ell - 1, (\lambda - 1)u\}.$$

Then the entire RH-relevant initial segment satisfies simultaneously

$$S_r = -\sigma \sum_{q\le\lambda u/2} c_q(u+r,u)M(q) > 0, \qquad 1 \le r \le h_\lambda(u),$$

and $|S_r - S_{r-1}| \le 1$. For $\lambda = 5/4$ this is the segment $r \le \min\{\ell - 1, u/4\}$ and all dependencies lie below $5u/8$.

Definition 8.47 (Relaxed divisor-sum window class)**.** Fix $u \geq 2$ and $1 \leq h < u$. For $\delta = (\delta_d)_{d \leq u} \in \{-1, 0, 1\}^u$ with $\delta_1 = 1$ and $\sum_{d \leq u} \delta_d = 0$, put

$$D_\delta(n) := \sum_{\substack{d \mid n \\ d \leq u}} \delta_d, \qquad u < n \leq u + h.$$

We call δ window-admissible with orientation $\sigma \in \{-1, 1\}$ if $D_\delta(n) \in \{-1, 0, 1\}$ throughout the window and

$$\sigma \sum_{u < n \leq u + r} D_\delta(n) < 0 \qquad (1 \leq r \leq h).$$

The induced oriented path is $S_r = -\sigma \sum_{u < n \leq u + r} D_\delta(n) > 0$. For the genuine Möbius prefix, $\delta_d = \mu(d)$, these are exactly the increment and excursion conditions coming from $\mu * 1 = \delta_1$.

Proposition 8.48 (Ballistic counterexample to the relaxed window class)**.** *The relaxed class of Definition 8.47 admits ballistic paths. Namely, take*

$$\delta_1 = 1, \qquad \delta_u = -1, \qquad \delta_d = 0 \quad (1 < d < u).$$

Then for every $u < n < 2u$ one has $D_\delta(n) = 1$, and with $\sigma = -1$,

$$S_r = r \qquad (1 \leq r < u).$$

If $m_q := \sum_{d \leq q} \delta_d$, then $m_q = 1$ for every $q < u$. Hence for any $0 < \beta < 1$, any fixed $1 < \lambda < 2$, and $h = \lfloor (\lambda - 1)u \rfloor$,

$$\sup_{q \leq \lambda u/2} \frac{|m_q|}{q^\beta} = 1, \qquad \frac{S_h}{(u + h)^\beta} \asymp_\lambda u^{1-\beta} \longrightarrow \infty.$$

Thus a maximum principle comparing the normalized localized-window height to the normalized history is false on this relaxed class.

Proof. For $u < n < 2u$, the integer u does not divide n. Hence among the two nonzero coordinates of δ only $d = 1$ contributes to $D_\delta(n)$, so $D_\delta(n) = 1$. The remaining assertions are immediate. □

Remark 8.49 (What the relaxation loses). The counterexample violates the internal divisor identities of the Möbius prefix. Already for every prime $p < u$ the genuine identity requires $\delta_1 + \delta_p = 0$, whereas the ballistic vector above has $\delta_p = 0$. Consequently any successful localized argument of this type must use some of the relations

$$\sum_{d \mid n} \delta_d = 0 \qquad (2 \leq n \leq u),$$

not merely the future-window constraints.

Remark 8.50 (The prefix correlation identity). The complete-excursion bridge identity must not be applied to a truncated localized window. For $1 \leq h < \ell$, put $x_j = \sigma \mu(u + j)$, $S_r = \sum_{j \leq r} x_j$, and $C_a^{[h]} = \sum_{j=1}^{h-a} x_j x_{j+a}$. Lemma 4.1 applied to the prefix of length h gives exactly

$$\sum_{r=1}^{h-1} S_r^2 + \sum_{a=1}^{h-1} a C_a^{[h]} = S_h \sum_{r=1}^{h-1} S_r.$$

The right-hand side vanishes only for a complete bridge. Thus $Q_\nu = -\sum_a a C_a^{(\nu)}$ cannot be used as an identity internal to a truncated horizon unless the excursion actually returns there. In particular, passing from a large localized value to the energy of the whole excursion mixes the retained window with the part deliberately discarded by the horizon theorem.

Remark 8.51 (The one-step iteration gap). Even a one-step implication sending a large value in the localized window to a large value at some $q \le \lambda u/2$ would not automatically iterate. The point q need not lie in the localized window of the excursion that contains q; it may lie beyond $\lambda u'$ relative to that excursion's start u'. Sparsity of the family of λ-long excursions does not repair a single descent chain. Iteration therefore requires either a principle valid for all t, or a mechanism that places the new point inside an admissible window at every step.

Proposition 8.52 (Termwise absolute-value contraction gives no nontrivial gain)**.** *Let $(u, x_1]$ be an excursion, $1 \le h < \min\{x_1 - u, u\}$, and let $w_1, \dots, w_h \ge 0$, not all zero. Put $K_q = \sum_{r \le h} w_r c_q(u + r, u)$, so that $\sum_r w_r M(u + r) = -\sum_q K_q M(q)$. If, for some $0 < \beta < 1$ and $C_\beta > 0$,*

$$\sum_q |K_q| q^\beta \le C_\beta \sum_{r \le h} w_r (u + r)^\beta,$$

then

$$\sum_{r \le h} w_r |M(u + r)| \le (1 + 2C_\beta) \sum_{r \le h} w_r (u + r)^\beta.$$

Proof. Proposition 8.44(iii) gives $c_1(u + r, u) \ge (r - 1)/2 \ge 0$, hence $\frac{1}{2} \sum_r w_r (r - 1) \le K_1 \le C_\beta \sum_r w_r (u+r)^\beta$. Since $M(u) = 0$ and M has unit steps, $|M(u+r)| \le r$, and the stated inequality follows. □

Remark 8.53 (No strict coercivity from the first coefficient). Writing $t = u + r$ gives the exact formula

$$c_1(u + r, u) = \begin{cases} \lceil r/2 \rceil, & u \text{ even}, \\ \lfloor r/2 \rfloor, & u \text{ odd}. \end{cases}$$

In particular $c_1(u + 1, u) = 0$ when u is odd. The preceding proposition therefore rules out only this particular termwise absolute-value mechanism; it does not rule out a global signed contraction or cancellation among the coefficients K_q.

Definition 8.54 (Multiplicative layers of the prefix)**.** Let $\Omega(n)$ denote the number of prime factors of n, counted with multiplicity. For $R \ge 2$ and $k \ge 1$, a vector $(\delta_d)_{d \le R}$ satisfies the first k Möbius layers if $\delta_1 = 1$ and

$$\sum_{d | n} \delta_d = 0 \qquad (2 \le n \le R,\ \Omega(n) \le k).$$

Thus layer 1 imposes the identities at primes, layer 2 adds prime squares and products of two primes, and so on.

Proposition 8.55 (One-layer propagation into the localized future)**.** *Suppose $(\delta_d)_{d \le R}$ satisfies the first k Möbius layers. Then*

$$\delta_d = \mu(d) \qquad (d \le R,\ \Omega(d) \le k).$$

Moreover, for every integer n with

$$R < n \le 2R, \qquad \Omega(n) \le k + 1,$$

one has

$$-\sum_{\substack{d | n \\ d \le R}} \delta_d = \mu(n).$$

Thus the k-th multiplicative layer of the prefix determines exactly the $(k+1)$-st multiplicative layer of the future increments in a window of length at most R.

Proof. The first assertion follows recursively from $\delta_n = -\sum_{d | n,\ d < n} \delta_d$: every proper divisor of an integer with $\Omega(n) \le k$ has smaller Ω. For the second, every proper divisor of $n \le 2R$ is at most $n/2 \le R$ and has $\Omega(d) \le k$. Hence the displayed sum equals $-\sum_{d | n,\ d < n} \mu(d) = \mu(n)$. □

Question 8.56 (Multiplicative-layer depth)**.** Fix $\theta \in (\theta_*, 1)$ and the localized excursion window $u < n \le 2\theta u$. How many internal Möbius layers below $R = \theta u$ are needed, together with $M(u) = 0$ and the excursion-window constraints, to exclude a height larger than $u^{1/2+\varepsilon}$? A bounded number of layers would give a genuinely compressed arithmetic mechanism; requiring all layers would amount to reconstructing the full Möbius prefix. Proposition 8.55 makes this depth a quantifiable intermediate invariant.

Proposition 8.57 (Layer sums in the localized window)**.** *Fix $1 < \lambda < 2$ and $j \ge 1$. As $u \to \infty$,*

$$\sum_{\substack{u<n\le \lambda u \\ \Omega(n)=j}} \mu(n) = (-1)^j\,(1+o(1))\,\frac{(\lambda-1)\,u\,(\log\log u)^{j-1}}{(j-1)!\,\log u}. \tag{8.25}$$

Proof. If $\Omega(n) = j$ and $\mu(n) \neq 0$, then n is squarefree with exactly j prime factors and $\mu(n) = (-1)^j$. By Landau's theorem (see [12, Ch. 7]), the number $\pi_j^\flat(x)$ of such $n \le x$ satisfies $\pi_j^\flat(x) \sim f_j(x) := x(\log\log x)^{j-1}/((j-1)!\log x)$. Since $f_j(x)/x$ is slowly varying, $f_j(\lambda u) - f_j(u) = (\lambda - 1 + o(1))f_j(u)$. □

Corollary 8.58 (Bounded layer depth)**.** *Fix $k \ge 1$, $\theta \in (\theta_*, 1)$, $R = \theta u$, and the window $u < n \le 2\theta u$.*

(i) *The increments fixed by the first k layers (Proposition 8.55), namely those with $\Omega(n) \le k+1$, form a set of $o(u)$ integers of the window.*

(ii) *Their summed contribution over the window is*

$$\sum_{\substack{u<n\le 2\theta u \\ \Omega(n)\le k+1}} \mu(n) = (-1)^{k+1}(1+o(1))\frac{(2\theta-1)\,u\,(\log\log u)^k}{k!\,\log u},$$

so it exceeds $u^{1-\varepsilon}$ in modulus and has a fixed sign.

(iii) *Let $\delta_d = \mu(d)$ for $\Omega(d) \le k$ and $\delta_d = 0$ otherwise. For squarefree n in the window with $\omega(n) = w \ge k+2$,*

$$D_\delta(n) = \sum_{\substack{d\mid n \\ d\le R}} \delta_d = \sum_{j=0}^{k} (-1)^j \binom{w}{j} = (-1)^k \binom{w-1}{k},$$

which has modulus at least $k+1 \ge 2$. Thus the zero extension of a k-layer prefix violates the unit-step constraint at every such n, i.e. at all but $o(u)$ squarefree integers of the window.

Proof. (i) and (ii) follow from Proposition 8.57 summed over $j \le k+1$, the term $j = k+1$ dominating. For (iii), all divisors of n other than n are $\le n/2 \le R$, and n itself has $\Omega(n) = w > k$, so $D_\delta(n)$ is the truncated alternating binomial sum, evaluated by induction on k via $\binom{w-1}{k} = \binom{w}{k} - \binom{w-1}{k-1}$. The proportion of squarefree $n \le x$ with $\omega(n) \le k+1$ tends to 0 by Proposition 8.57. □

Remark 8.59 (What bounded depth can and cannot do). Corollary 8.58 settles the determinacy side of the layer question. A bounded number of layers fixes only a density-zero set of future increments, and that set alone pushes M by an amount of order $u(\log\log u)^k/\log u$ with a fixed sign, far beyond $u^{1/2+\varepsilon}$. The RH scale is therefore reached only through cancellation among the $\asymp \log\log u$ layers carrying most integers, as in the large-prime split of Proposition 10.13. On the other hand, part (iii) shows that the unit-step constraint is far from vacuous on a k-layer relaxation: it forbids the naive completion at almost every integer of the window, so the free coordinates δ_d with $\Omega(d) > k$ are strongly constrained. Whether these constraints alone exclude an excursion of height $u^{1/2+\varepsilon}$ is the remaining content of Question 8.56.

8.6. **Conditional extreme-value consequence of excursion sparsity.** The counts of Section 8.1 bound the number of excursions in a family, not their heights. They acquire a second function when combined with a tail hypothesis on a normalized excursion statistic: they fix the scale of the family maximum, with no independence assumption.

Let $(\Omega, \mathcal{G}, \Pr)$ be a probability space carrying a random sequence of disjoint integer intervals $I_k = (x_{0,k}, x_{1,k}]$ with $x_{1,k} \le x_{0,k+1}$, and real random variables Z_k (for instance a model of the Möbius walk with $Z_k = H_k/\sqrt{\ell_k}$). A family $\mathcal{F}$ is given by a rule selecting indices from $(x_{0,k}, x_{1,k})_k$, such as $\mathcal{F}_h$; let $k_{\mathcal{F}}(1) < k_{\mathcal{F}}(2) < \cdots$ enumerate its members, and put

$$Z_{\max,\mathcal{F}}(X) := \max\{Z_k : k \in \mathcal{F},\ x_{0,k} \le X\} \qquad (\max \emptyset = -\infty).$$

The lemmas of Section 8.1 hold for every ω.

Hypothesis 8.60 (SG_p)**.** There are $C, c, p > 0$ such that for every $j \ge 1$ and $z \ge 0$

$$\Pr\big(k_{\mathcal{F}}(j) \text{ exists and } Z_{k_{\mathcal{F}}(j)} > z\big) \le Ce^{-cz^p}.$$

For $p = 2$ this is a uniform sub-Gaussian tail, denoted (SG).

Theorem 8.61 (Extreme scale from a count)**.** *Assume* (SG_p) *for $\mathcal{F}$ and suppose that $E_{\mathcal{F}}(X) \le L(X)$ for every ω, with $L(X) \ge 1$ deterministic. Then*

$$\Pr\big(Z_{\max,\mathcal{F}}(X) > z\big) \le L(X)\, Ce^{-cz^p}. \tag{8.26}$$

Consequently, for every $\delta > 0$,

$$\Pr\Big(Z_{\max,\mathcal{F}}(X) > \Big(\frac{1+\delta}{c}\log L(X)\Big)^{1/p}\Big) \le C\, L(X)^{-\delta}, \tag{8.27}$$

and $Z_{\max,\mathcal{F}}(X) = O_{\Pr}\big((\log L(X))^{1/p}\big)$ as $L(X) \to \infty$.

Proof. The members of $\mathcal{F}$ starting at most X are the first $E_{\mathcal{F}}(X) \le L(X)$ members of the enumeration, so $\{Z_{\max,\mathcal{F}}(X) > z\} \subset \bigcup_{j \le L(X)} \{k_{\mathcal{F}}(j) \text{ exists},\ Z_{k_{\mathcal{F}}(j)} > z\}$, and (8.26) is the union bound. Substituting $cz^p = (1+\delta)\log L(X)$ gives (8.27). □

Corollary 8.62 (Triple-logarithmic scale on the square family)**.** *Assume (SG) for the square-exceptional family $\mathcal{F}_{x^2}$ of the model. Then, for $X \ge e^e$ and every $\delta > 0$,*

$$\Pr\big(Z_{\max,\square}(X) > z\big) \le 4C(\log\log X)e^{-cz^2}, \tag{8.28}$$

$$\Pr\Big(Z_{\max,\square}(X) > \sqrt{\tfrac{1+\delta}{c}\log\log\log X}\Big) \le 4C(\log\log X)^{-\delta}. \tag{8.29}$$

Thus $Z_{\max,\square}(X) = O_{\Pr}(\sqrt{\log\log\log X})$. The same holds for the family $\mathcal{F}_{cx^2}$ of Lemma 8.2 with a constant depending on c.

Proof. Use (8.2) in (8.26) and note that $e^{-cz^2} = (\log\log X)^{-(1+\delta)}$ at the stated z. □

The union bound gives convergence in probability; a Borel–Cantelli argument along a sequence adapted to L gives an almost sure bound with the same constant.

Theorem 8.63 (Almost sure extreme scale)**.** *Assume* (SG_p) *for $\mathcal{F}$ and $E_{\mathcal{F}}(X) \le L(X)$ for every ω, where L is continuous, nondecreasing and unbounded. Then almost surely*

$$\limsup_{X\to\infty} \frac{Z_{\max,\mathcal{F}}(X)}{(\log L(X))^{1/p}} \le c^{-1/p}.$$

Proof. Fix $\delta > 0$. For $j \ge j_0$ let X_j be the least X with $L(X) = 2^j$. Put $z_j = \big((1+\delta)(j+1)\log 2/c\big)^{1/p}$ and $A_j = \{Z_{\max,\mathcal{F}}(X_{j+1}) > z_j\}$. By (8.26), $\Pr(A_j) \le 2^{j+1}Ce^{-(1+\delta)(j+1)\log 2} = C2^{-(j+1)\delta}$, which is summable, so almost surely A_j occurs only finitely often. For $X_j \le X \le X_{j+1}$, monotonicity of $Z_{\max,\mathcal{F}}$ in X gives $Z_{\max,\mathcal{F}}(X) \le z_j$ outside A_j, while $\log L(X) \ge j\log 2$. Hence

the lim sup is at most $\lim_j\big((1+\delta)(j+1)/(cj)\big)^{1/p} = ((1+\delta)/c)^{1/p}$. Let $\delta \downarrow 0$ along a countable sequence. □

With $L_\square(X) = 2 + \log_2(\log X/\log 2)$, which is continuous and increasing, Theorem 8.63 gives under (SG)

$$\limsup_{X\to\infty} \frac{Z_{\max,\square}(X)}{\sqrt{\log\log\log X}} \le c^{-1/2} \quad \text{almost surely.}$$

Applied to the three counts available for any sequence of disjoint intervals (all excursions: at most X; Lemma 8.4; Lemma 8.1), Theorems 8.61 and 8.63 give three levels:

family	members starting $\le X$	extreme scale under (SG)
all excursions	$\le X$	$\sqrt{\log X}$
$x_1 > \lambda x_0$	$O(\log X)$	$\sqrt{\log\log X}$
$x_1 > c x_0^2$ (or x_0^A)	$O(\log\log X)$	$\sqrt{\log\log\log X}$

Each level of lacunarity removes one logarithm from the extreme scale. Under (SG_p) the last column reads $(\log X)^{1/p}$, $(\log\log X)^{1/p}$, $(\log\log\log X)^{1/p}$.

Remark 8.64 (Scope). For M itself the square-exceptional family is finite by Corollary 8.8, so Theorems 8.61–8.63 concern stochastic models of the Möbius walk, in which the pathwise counts of Section 8.1 persist while no analogue of the Kaczorowski–Pintz localization is available. Hypothesis (SG) is an assumption on the model; the theorems state exactly what the counts contribute to its extremes.

8.7. **The square range as an exact bilinear form of the prefix.**

Proposition 8.65 (Bilinear square-range identity)**.** *Let $u \ge 1$ and $1 \le t \le u^2$. Then*

$$M(t) = 2M(\min(t,u)) - \sum_{d,j\le u} \mu(d)\mu(j)\left\lfloor \frac{t}{dj}\right\rfloor. \tag{8.30}$$

In particular, if $M(u) = 0$ and $u < t \le u^2$,

$$M(t) = -\mathcal{B}_u(t), \qquad \mathcal{B}_u(t) := \sum_{d,j\le u} \mu(d)\mu(j)\left\lfloor \frac{t}{dj}\right\rfloor, \qquad \mathcal{B}_u(u) = 0. \tag{8.31}$$

Proof. Let $\mu_u = \mu\mathbf{1}_{[1,u]}$. Each factor of $(\mu-\mu_u)*(\mu-\mu_u)$ is supported on integers $> u$, so the convolution, and its convolution with 1, vanish on $n \le u^2$. Expanding and using $\mu * 1 = \delta$ gives $\mu(n) = 2\mu_u(n) - (\mu_u * \mu_u * 1)(n)$ for $n \le u^2$; sum over $n \le t$. □

Remark 8.66 (Higher-order determinacy). For every $k \ge 1$ and $n \le u^k$,

$$\mu(n) = \sum_{j=1}^{k} (-1)^{j-1}\binom{k}{j}\big(\mu_u^{*j} * 1^{*(j-1)}\big)(n), \tag{8.32}$$

because $(\mu-\mu_u)^{*k} * 1^{*(k-1)}$ vanishes on $n \le u^k$; expand by the binomial theorem and use $\mu*1=\delta$. Thus determination by the prefix is not special to u^2. What is special to $t \le u^2$ is the order-two form (8.31), with rank-one coefficients $\mu(d)\mu(j)$ and elementary kernel $\lfloor t/(dj)\rfloor$. This is a truncation identity of Heath-Brown/Vaughan type [4]; summed forms of this kind are used in the fast computation of M [5].

Question 8.67 (Square-excursion bridges)**.** Let $u = x_{0,\nu}$ and $x_{1,\nu} \le u^2$. Then $\mathcal{B}_u(u) = \mathcal{B}_u(x_{1,\nu}) = 0$ and $-\sigma_\nu \mathcal{B}_u(t) > 0$ for $u < t < x_{1,\nu}$. Can the rank-one coefficients $\mu(d)\mu(j)$, the kernel $\lfloor t/(dj)\rfloor$ and these bridge and sign constraints be combined to prove

$$\max_{u<t\le x_{1,\nu}} |\mathcal{B}_u(t)| \ll_\varepsilon x_{1,\nu}^{1/2+\varepsilon}$$

uniformly over square-ordinary excursions? By Corollary 8.12 this uniform bound is equivalent to RH. By Corollary 8.43 it suffices to consider $t \le \min\{x_{1,\nu}, \frac{5}{4}u\}$, where $\mathcal{B}_u(t)$ reduces to the localized linear form and uses only $\mu(d)$ with $d \le 5u/8$.

8.8. **Mellin mass of a complete excursion.**

Proposition 8.68 (Excursion blocks of $1/\zeta$)**.** *For a complete excursion $I = (x_0, x_1]$ put $D_I(s) = \sum_{x_0<n\le x_1} \mu(n)n^{-s}$. Then*

$$D_I(s) = \sum_{n=x_0+1}^{x_1-1} M(n)\big(n^{-s} - (n+1)^{-s}\big). \tag{8.33}$$

For real $\sigma > 0$, $D_I(\sigma)$ has the sign of the excursion and

$$|D_I(\sigma)| \ge \sigma\beta_I x_1^{-\sigma-1} \ge \sigma H_I^2 x_1^{-\sigma-1}. \tag{8.34}$$

For $\Re s > 1$,

$$\frac{1}{\zeta(s)} = \sum_I D_I(s), \tag{8.35}$$

zero runs contributing nothing.

Proof. Summation by parts gives (8.33) because M vanishes at both endpoints. All interior $M(n)$ have the sign of the excursion, and $n^{-\sigma} - (n+1)^{-\sigma} \ge \sigma x_1^{-\sigma-1}$ for $n < x_1$. The bound $\beta_I \ge H_I^2$ is (4.13). If M vanishes at two consecutive integers the intervening $\mu(n)$ is zero, so every n with $\mu(n) \ne 0$ lies in an excursion; sum the disjoint blocks in the half-plane of absolute convergence. □

9. Trial-division horizon of an excursion

Lemma 9.1 (Square horizon)**.** *Let $u \ge 2$. Trial division by the primes $p \le u$ alone determines $\mu(n)$ for every $n \le u^2$.*

Proof. Remove the prime factors $p \le u$ of n; if one occurs twice, $\mu(n) = 0$. The cofactor $q \le u^2$ has all prime factors $> u$, so it is 1 or a single prime. □

Remark 9.2 (Determinacy is not information). Primes $> u$ are not logically new information, and determination by a finite prefix is not special to u^2: (8.32) propagates a prefix to every power range, and (8.21) determines M from $M(1)$. The ranges $(u, 2u]$ and $(u, u^2]$ are special only in the concrete sense that the exact summatory formula is of order one ((8.19)) or two ((8.31)) in the prefix, and that one-stage trial division by primes $\le u$ suffices on $(u, u^2]$.

Definition 9.3 (Smooth and rough parts)**.** For $y > 1$ write $n = s_y(n)\, r_y(n)$, where every prime factor of $s_y(n)$ is $\le y$ and every prime factor of $r_y(n)$ is $> y$. Let $P^+(n)$ be the largest prime factor of n, with $P^+(1) = 1$.

Lemma 9.4. *$\mu(n) = \mu(s_y(n))\,\mu(r_y(n))$. If $n < y^k$ then $\Omega(r_y(n)) \le k-1$, and if $n \le y^2$ then $r_y(n)$ is 1 or a prime.*

Proof. The two parts are coprime. If $\Omega(r) \ge 1$ then $y^{\Omega(r)} < r \le n$. □

Remark 9.5 (The horizon u^2 is not the multiplicative threshold of an excursion). For a square-ordinary excursion $\sqrt{v_\nu} \le u_\nu$. Every $n \le v_\nu$ has at most one prime factor $> \sqrt{v_\nu}$, so the threshold that governs the multiplicative structure inside I_ν is $\sqrt{v_\nu}$, below u_ν. The regime $t > u_\nu^2$ of Section 10 lies beyond v_ν for a square-ordinary excursion; that section concerns $M(t)$ globally, and Section 11 gives the local version inside an excursion.

10. Large-prime decomposition at level u^2

The identities of this section are decompositions by large prime factors of Buchstab/Legendre type. The asymptotic statements use only the prime number theorem with classical error term and the Vinogradov–Korobov bound [12, 19].

Definition 10.1 (Smooth baseline and large-prime part)**.** For $u \geq 2$ and $t \geq 1$ put

$$S_u(t) := \sum_{\substack{n \leq t \\ P^+(n) \leq u^2}} \mu(n), \qquad E_u(t) := \sum_{\substack{n \leq t \\ P^+(n) > u^2}} \mu(n), \qquad M(t) = S_u(t) + E_u(t). \tag{10.1}$$

Proposition 10.2 (Single-prime copy)**.** *For a prime p and $t < p^2$,*

$$\Delta_p(t) := \sum_{a \leq t/p} \mu(ap) = -M(t/p). \tag{10.2}$$

Proof. $a \leq t/p < p$, so $p \nmid a$ and $\mu(ap) = -\mu(a)$. □

Proposition 10.3 (Sign content of one prime cone)**.** *Let $\mathcal{S}(y) = \sum_{a \leq y} \mu(a)^2$. Among the numbers ap with $a \leq y < p$, the numbers of values $+1, -1, 0$ of μ are*

$$N_+ = \frac{\mathcal{S}(y) - M(y)}{2}, \qquad N_- = \frac{\mathcal{S}(y) + M(y)}{2}, \qquad N_0 = \lfloor y \rfloor - \mathcal{S}(y), \tag{10.3}$$

with limiting proportions $3/\pi^2,\ 3/\pi^2,\ 1 - 6/\pi^2$.

Proof. $\mu(ap) = -\mu(a)$, and $\#\{a \leq y : \mu(a) = \mp 1\} = (\mathcal{S}(y) \mp M(y))/2$. □

Proposition 10.4 (First-order window)**.** *If $u^2 < t < u^4$, no $n \leq t$ has two prime factors $> u^2$ (with multiplicity), and*

$$E_u(t) = -\sum_{u^2 < p \leq t} M(t/p). \tag{10.4}$$

Proof. Two such factors would give $n > u^4 > t$. Hence $E_u(t) = \sum_{u^2 < p \leq t} \Delta_p(t)$, and $t < u^4 < p^2$; apply (10.2). □

Proposition 10.5 (Prime-count form)**.** *For $u^2 < t < u^4$ and $R := t/u^2$,*

$$E_u(t) = -\sum_{a \leq R} \mu(a)\big(\pi(t/a) - \pi(u^2)\big), \qquad M(t) = S_u(t) - \sum_{a \leq R} \mu(a)\pi(t/a) + M(R)\pi(u^2). \tag{10.5}$$

Proof. Expand (10.4) as $-\sum_p \sum_{a \leq t/p} \mu(a)$ and interchange: for fixed a the prime runs over $u^2 < p \leq t/a$, which is non-empty only if $a < R$. □

Thus the only input from primes $> u^2$ is the value of π at the points t/a, $a \leq R$, weighted by $\mu(a)$ at the small scale $R < u^2$.

Definition 10.6 (Copy multiplicities)**.** For $u^2 < t < u^4$, $R = t/u^2$ and $1 \leq y \leq \lfloor R \rfloor$ put

$$N_y(t,u) = \#\Big\{p :\ \max\big(u^2, \tfrac{t}{y+1}\big) < p \leq \tfrac{t}{y}\Big\}, \qquad \text{so that} \qquad E_u(t) = -\sum_{1 \leq y \leq \lfloor R \rfloor} N_y(t,u) M(y). \tag{10.6}$$

Remark 10.7 (Asymptotics of the multiplicities). For fixed $R > 1$ and $u \to \infty$,

$$N_y \sim \frac{Ru^2}{2\log u} \cdot \frac{1}{y(y+1)} \quad (1 \leq y \leq \lfloor R \rfloor - 1), \qquad N_{\lfloor R \rfloor} \sim \frac{u^2}{2\log u}\Big(\frac{R}{\lfloor R \rfloor} - 1\Big) \quad (R \notin \mathbb{Z}), \tag{10.7}$$

and $N_R = 0$ if $R \in \mathbb{Z}$. The top multiplicity is truncated because $t/(\lfloor R \rfloor + 1) < u^2$.

Definition 10.8 (Coherent coefficient)**.**

$$\mathcal{C}(R) := \int_1^R \frac{M(w)}{w^2}\,dw = \sum_{a\le R}\mu(a)\Big(\frac{1}{a}-\frac{1}{R}\Big) = \sum_{a\le R}\frac{\mu(a)}{a} - \frac{M(R)}{R}. \tag{10.8}$$

Proposition 10.9 (Fixed R)**.** *For fixed $R>1$, as $u\to\infty$,*

$$E_u(Ru^2) = -\frac{Ru^2}{2\log u}\big(\mathcal{C}(R)+o(1)\big). \tag{10.9}$$

Proof. In (10.5) the sum is finite; for $a<R$ the prime number theorem gives $\pi(Ru^2/a)-\pi(u^2)\sim (R/a-1)u^2/(2\log u)$, and $\sum_{a\le R}\mu(a)(R/a-1)=R\mathcal{C}(R)$. □

Lemma 10.10 (The multiplicity-weighted coefficient)**.** *The coefficient*

$$\Theta(R) := \sum_{1\le y\le R}\frac{M(y)}{y(y+1)} = \sum_{n\le R}\frac{\mu(n)}{n} - \frac{M(R)}{\lfloor R\rfloor+1} \tag{10.10}$$

satisfies

$$\Theta(R) = \mathcal{C}\big(\lfloor R\rfloor+1\big), \qquad \mathcal{C}(R) = \Theta(R) - M(R)\Big(\frac{1}{R}-\frac{1}{\lfloor R\rfloor+1}\Big), \tag{10.11}$$

and

$$\mathcal{C}(R) = -\int_R^\infty \frac{M(w)}{w^2}\,dw, \qquad \Theta(R) = -\sum_{n>R}\frac{M(n)}{n(n+1)}. \tag{10.12}$$

Proof. M is constant on $[y,y+1)$ and $\int_y^{y+1} w^{-2}dw = 1/(y(y+1))$, which gives (10.11). For $s>1$, (7.1) gives $\int_1^\infty M(w)w^{-s-1}\,dw = 1/(s\zeta(s))\to 0$ as $s\downarrow 1$. For $s\in[1,1+\delta]$ the bound $M(w)\ll we^{-c\sqrt{\log w}}$ dominates the integrand by an integrable multiple of $e^{-c\sqrt{\log w}}/w$, so dominated convergence gives $\int_1^\infty M(w)w^{-2}dw=0$. □

Remark 10.11 ($\mathcal{C}$ versus Θ). Summing (10.7) against $M(y)$ suggests the coefficient $\Theta(R)$, but the truncated top multiplicity replaces it by $\mathcal{C}(R)$; by (10.11) the two agree exactly when $M(\lfloor R\rfloor)=0$. For $1<R<2$ one has $E_u=-(\pi(t)-\pi(u^2))\sim -(R-1)u^2/(2\log u)$, i.e. $\mathcal{C}(R)=1-1/R$, whereas $\Theta(R)=1/2$.

Corollary 10.12 (Decay of the coefficients)**.** *By* (10.12) *and* (6.2)*,*

$$\mathcal{C}(R),\ \Theta(R) \ll \exp\big(-c_1(\log R)^{3/5}(\log\log R)^{-1/5}\big). \tag{10.13}$$

Proposition 10.13 (Both parts are large)**.** (i) *For fixed $R>1$ with $\mathcal{C}(R)\neq 0$,*

$$E_u(Ru^2)\sim -\frac{Ru^2}{2\log u}\mathcal{C}(R), \qquad S_u(Ru^2)\sim +\frac{Ru^2}{2\log u}\mathcal{C}(R); \tag{10.14}$$

for $1<R<2$, $\mathcal{C}(R)=1-1/R>0$.

(ii) *There is $C>0$ such that, as $t\to\infty$, uniformly for $u^2<t<u^4$ with $R=t/u^2\ge \exp(C(\log\log t)^2)$,*

$$E_u(t) = (1+o(1))\frac{t}{(\log t)^2}, \qquad S_u(t) = -(1+o(1))\frac{t}{(\log t)^2}. \tag{10.15}$$

In particular, in both regimes $|E_u(t)|$ and $|S_u(t)|$ exceed $t^{1/2+\varepsilon}$ for every fixed $\varepsilon<1/2$ and all large t, and only their sum M can be compared with the scale $t^{1/2+\varepsilon}$.

Proof. We use the classical form [12]: for some $c > 0$,

$$\pi(y) = \mathrm{li}(y) + O\big(ye^{-c\sqrt{\log y}}\big), \quad M(y) \ll ye^{-c\sqrt{\log y}}, \quad \widetilde{m}(y) := \sum_{a\le y} \frac{\mu(a)}{a} \ll e^{-c\sqrt{\log y}}, \tag{10.16}$$

and $\sum_{n\ge1} \mu(n)\log n/n = -1$ as a convergent series (the value at $s = 1$ of $-\frac{d}{ds}\zeta(s)^{-1}$, since $\zeta'/\zeta^2 \to -1$). The integral below converges absolutely by the bound on $\widetilde{m}$, and

$$\int_1^\infty \widetilde{m}(w)\frac{dw}{w} = \lim_{Y\to\infty}\Big(\widetilde{m}(Y)\log Y - \sum_{a\le Y}\frac{\mu(a)\log a}{a}\Big) = 1. \tag{10.17}$$

(i) Proposition 10.9, and $M(t) = o(t/\log t)$. For $1 < R < 2$, $M = 1$ on $[1, 2)$.

(ii) Since $t < u^4$, $R < \sqrt{t}$, so $\log(t/a) \ge \frac{1}{2}\log t$ for $a \le R$. By (10.5),

$$E_u(t) = -\sum_{a\le R}\mu(a)\,\mathrm{li}(t/a) - \sum_{a\le R}\mu(a)\big(\pi(t/a) - \mathrm{li}(t/a)\big) + M(R)\pi(u^2). \tag{10.18}$$

The second term is $\ll t(1+\log R)e^{-c\sqrt{\log t/2}} = o(t/\log^2 t)$. For the third, $\frac{1}{4}\log t < \log u < \frac{1}{2}\log t$, so

$$|M(R)|\pi(u^2) \ll Re^{-c\sqrt{\log R}}\frac{u^2}{\log u} \ll e^{-c\sqrt{\log R}}\frac{t}{\log t} = o(t/\log^2 t)$$

once $c\sqrt{C} > 1$. For the first, with $F(w) = w\,\mathrm{li}(t/w)$,

$$\sum_{a\le R}\mu(a)\,\mathrm{li}(t/a) = \widetilde{m}(R)F(R) - \int_1^R \widetilde{m}(w)F'(w)\,dw, \tag{10.19}$$

$$F'(w) = \mathrm{li}(t/w) - \frac{t/w}{\log(t/w)} = \frac{t/w}{\log^2(t/w)}\Big(1 + O\Big(\frac{1}{\log t}\Big)\Big).$$

The boundary term is $\ll e^{-c\sqrt{\log R}}t/\log t = o(t/\log^2 t)$. The integral equals

$$\frac{t}{\log^2 t}\left(\int_1^R \widetilde{m}(w)g_t(w)\frac{dw}{w} + O(1/\log t)\right), \qquad g_t(w) = \frac{\log^2 t}{\log^2(t/w)} \in [1, 4].$$

Extend $g_t(w)\mathbf{1}_{\{w\le R\}}$ by zero to $[1,\infty)$. Since $R \ge \exp(C(\log\log t)^2) \to \infty$, this factor tends to 1 pointwise and is bounded by 4, while $|\widetilde{m}(w)|/w$ is integrable; dominated convergence gives $\int_1^R \widetilde{m}(w)g_t(w)\,dw/w = 1 + o(1)$, where the $o(1)$ depends only on t and tends to 0 as $t \to \infty$ uniformly over all pairs (u, R) in the stated range: indeed $g_t(w)\mathbf{1}_{\{w\le R\}} \to 1$ uniformly for w in compact sets, since $R \ge \exp(C(\log\log t)^2)$ and $w \le W$ give $|g_t(w) - 1| \ll \log W/\log t$. Hence $E_u(t) = (1 + o(1))t/\log^2 t$, and $S_u = M - E_u$ with $M(t) = o(t/\log^2 t)$. □

Remark 10.14 (Why $t/\log^2 t$). $\mathcal{C}(R)$ captures only the $1/\log t$ term, whose coefficient tends to zero by Corollary 10.12. The $1/\log^2 t$ term comes from $\zeta'(s)/\zeta(s)^2 \to -1$, that is, from the pole of ζ at $s = 1$, and does not decay in R.

Remark 10.15 (First copy transitions). For $1 < R < 2$ every copy has length 1 and $E_u(t) = -(\pi(t) - \pi(u^2))$. For $2 < R < 3$ copies of length 2 appear but $M(2) = 0$. For $3 < R < 4$ copies of length 3 enter with $M(3) = -1$, of opposite net sign.

11. Local large-prime decomposition of an excursion

Proposition 11.1 (Local decomposition). *Let $I_\nu = (u, v]$ be a complete excursion and $y = \sqrt{v}$. For $u < t \le v$,*

$$M(t) = \sum_{\substack{u<n\le t\\ P^+(n)\le y}}\mu(n) - \sum_{a<y}\mu(a)\,\Pi_a(t), \qquad \Pi_a(t) := \#\Big\{p:\ \max\big(y, \tfrac{u}{a}\big) < p \le \tfrac{t}{a}\Big\}, \tag{11.1}$$

and the bridge condition $M(v) = 0$ reads

$$\sum_{\substack{u<n\le v\\ P^+(n)\le\sqrt{v}}} \mu(n) = \sum_{a<\sqrt{v}} \mu(a)\,\Pi_a(v). \tag{11.2}$$

Proof. $M(u) = 0$. An $n \le v = y^2$ has at most one prime factor $p > y$; then $n = ap$ with $a < v/y = y < p$, so $\mu(n) = -\mu(a)$, and $(a,p) \mapsto ap$ is a bijection onto these n. □

Remark 11.2 (Primes in short intervals). The decomposition holds for every excursion, with threshold $\sqrt{v_\nu}$. The large-prime input consists of prime counts in the intervals $(u/a, t/a]$, of length at most ℓ_ν/a at height about v_ν/a. Huxley proved the prime number theorem uniformly in intervals of length $h \ge x^{7/12+\varepsilon}$ [7]; the zero-density estimates of Guth–Maynard give intervals of length $x^{17/30+o(1)}$ [3]. For Möbius sums, Matomäki–Teräväinen prove $\sum_{x<n\le x+x^\theta} \mu(n) = o(x^\theta)$ for every $\theta > 0.55$ [11]. Below these ranges no asymptotic for Π_a is available uniformly in a.

Lemma 11.3 (Inverse logarithms)**.** *Let C be as in Proposition 10.13. Uniformly for*

$$\exp\bigl(C(\log\log x)^2\bigr) \le z \le \sqrt{x},$$

$$\sum_{a\le z} \frac{\mu(a)}{a\log(x/a)} = -\frac{1+o(1)}{(\log x)^2}. \tag{11.3}$$

Proof. Partial summation gives

$$\frac{\widetilde{m}(z)}{\log(x/z)} - \int_1^z \frac{\widetilde{m}(w)}{w\log^2(x/w)}\,dw.$$

Since $z \le \sqrt{x}$, the boundary term is $O(e^{-c\sqrt{\log z}}/\log x) = o(1/\log^2 x)$ uniformly. Write $\log^{-2}(x/w) = g_x(w)\log^{-2} x$ with $1 \le g_x(w) \le 4$, extend $g_x\mathbf{1}_{\{w\le z\}}$ by zero, and apply dominated convergence with (10.17). □

Heuristic 11.4 (Opposite drifts)**.** Let $u \ge \sqrt{v}\exp(C(\log\log v)^2)$. Near $t \in I_\nu$ the counts Π_a grow for $a < t/\sqrt{v}$, at rate $1/(a\log(t/a))$ if primes near t/a have their expected local density uniformly in a. Over $(t, t+h]$ the large-prime part then changes by about $-h\sum_{a<t/\sqrt{v}} \mu(a)/(a\log(t/a)) = (1+o(1))h/\log^2 t$ by Lemma 11.3. Since $M(v) - M(u) = 0$, the $\sqrt{v}$-smooth part drifts at rate about $-1/\log^2 v$, independently of σ_ν. As

$$\frac{\ell_\nu}{\log^2 v_\nu} \gg \sqrt{\ell_\nu} \iff \ell_\nu \gg \log^4 v_\nu,$$

for $\ell_\nu \gg (\log v_\nu)^4$ both drift contributions exceed $\sqrt{\ell_\nu}$, and (11.2) is a balance between two opposite drifts: the local analogue of Proposition 10.13.

12. A one-frequency model for the black curve

Heuristic 12.1 (Scale and period)**.** Motivated by (5.1), model $M(x) \approx \sqrt{x}\,F(\log x)$ with F almost periodic, and retain only the lowest zero γ_1 (RH does not imply that its coefficient dominates). Within an excursion of length $\asymp x$ this gives $H_\nu \asymp \sqrt{x}$ and $\sqrt{H_\nu}\ell_\nu \asymp x^{5/4}$, so the black curve oscillates at scale

$$|\mathcal{A}(x)| \asymp x^{5/4}, \tag{12.1}$$

the scale of Corollary 6.4, with period $2\pi/\gamma_1$ in $\log x$, i.e. a factor $e^{2\pi/\gamma_1} \approx 1.5597$ in x. In this model consecutive zeros of M are separated by the factor $e^{\pi/\gamma_1} = \lambda_*$, the threshold of Corollary 8.13(iii): a model excursion is exactly λ_*-long.

13. Numerical data

The following census covers $n \le N = 455\,052\,511$, with $Z(N) = 87\,385$ excursions, recording $(u_\nu, \tau_\nu, v_\nu, H_\nu, \sigma_\nu)$ and the curves $\mathcal{A}_+, \mathcal{A}_-, \mathcal{A}$:

$$\mathcal{A}_+(N) = 6.3449 \times 10^9, \quad \mathcal{A}_-(N) = -7.0877 \times 10^9, \quad \mathcal{A}(N) = -7.4279 \times 10^8, \tag{13.1}$$

$$\max_\nu \frac{H_\nu}{\sqrt{\ell_\nu}} = 2.0743, \qquad \max_\nu \frac{\beta_\nu}{\ell_\nu^{3/2}} = 1.0214. \tag{13.2}$$

Since $|M(x)| \le 0.571\sqrt{x}$ for $33 \le x \le 10^{16}$ [6], and $|M(n)| \le \sqrt{n}$ for $n < 33$ with equality only at $n = 1$, the constant $c = 1$ is admissible in Lemma 6.1, which gives $\mathcal{A}_+(N) + |\mathcal{A}_-(N)| \le \frac{4}{5}N^{5/4} = 5.317 \times 10^{10}$ against the observed 1.3433×10^{10}, a ratio of 3.96. Heuristic 7.5 with $L = \sqrt{N}$ predicts $\#\{\ell_\nu > \sqrt{N}\} \asymp N^{1/4} \approx 146$. At this range $N^{0.05} \approx 2.7$, so factors x^ε are not separable from constants. Reproducing these values requires the excursion file, the area-integration convention of Proposition 3.2, and the terminal convention of Definition 2.5.

14. Open problems

(1) In the localized form of Corollary 8.43, bound

$$\sum_{d \le 5u/8} \mu(d)\big(\lfloor t/d \rfloor - \lfloor u/d \rfloor\big)$$

by $t^{1/2+\varepsilon}$ on $u < t \le \min\{x_1, 5u/4\}$, using the full simultaneous intra-excursion constraints rather than pointwise estimates.

(2) Quantify the multiscale persistence forced by Lemma 8.38. By Corollary 8.40 and Remark 8.41, channel descent iterates but loses the factor $2K/k^\beta$ per step; the problem is to show that along a descent chain from an RH-violating value the loss is eventually absent, or that the deep remainder R_K carries the value and can be bounded.

(3) Question 8.56 in the form left by Corollary 8.58: bounded depth fixes only $o(u)$ window increments, whose sum is of order $u(\log\log u)^k/\log u$; decide whether the unit-step and excursion constraints on the remaining free coordinates δ_d, $\Omega(d) > k$, already force the cancellation down to $u^{1/2+\varepsilon}$. The first case is $k = 1$ (primes fixed, $\delta_p = -1$).

(4) Can the threshold $\lambda_* = e^{\pi/\gamma_1}$ of Corollary 8.13 be lowered? By Remark 8.29 this requires information on the position of large values inside excursions, beyond the count (8.6).

(5) Whether (6.4) is equivalent to RH.

(6) Whether the pruned local condition (7.7) holds, and whether the bridge constraints (4.5), (4.6) admit a third independent relation involving the defect $\ell_\nu/2 - H_\nu \ge 0$.

(7) Establish (SG) for $Z = H/\sqrt{\ell}$ in the simple random walk model of the Möbius walk, and compute the extreme scale of Corollary 8.62 there; the Brownian-excursion limit suggests $c = 2$.

(8) Numerically: the number of excursions with $\ell_\nu > \sqrt{N}$ against $N^{1/4}$, the maxima of $H_\nu/\sqrt{\ell_\nu}$ and $\beta_\nu/\ell_\nu^{3/2}$ over them; the split (11.1) and the σ_ν-independence of Heuristic 11.4; the frequency of touching runs and of ballistic excursions (Heuristic 7.8); and the distribution of x_1/x_0, to be compared with λ_*.

References

[1] P. Carmona and N. Pétrélis, *Limit theorems for random walk excursion conditioned to enclose a typical area*, arXiv:1709.06448.

[2] D. Denisov, M. Kolb and V. Wachtel, *Local asymptotics for the area of random walk excursions*, J. London Math. Soc. **91** (2015), 495–513.

[3] L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, Ann. of Math. (2) **203** (2026), no. 2, 623–675, doi:10.4007/annals.2026.203.2.6.

[4] D. R. Heath-Brown, *Prime numbers in short intervals and a generalized Vaughan identity*, Canad. J. Math. **34** (1982), 1365–1377.
[5] H. A. Helfgott and L. Thompson, *Summing $\mu(n)$: a faster elementary algorithm*, Res. Number Theory **9** (2023), Paper No. 6.
[6] G. Hurst, *Computations of the Mertens function and improved bounds on the Mertens conjecture*, Math. Comp. **87** (2018), no. 310, 1013–1028.
[7] M. N. Huxley, *On the difference between consecutive primes*, Invent. Math. **15** (1972), 164–170.
[8] S. Janson, *Brownian excursion area, Wright's constants in graph enumeration, and other Brownian areas*, Probab. Surveys **4** (2007), 80–145.
[9] J. Kaczorowski and J. Pintz, *Oscillatory properties of arithmetical functions. I*, Acta Math. Hungar. **48** (1986), 173–185.
[10] T. Kotnik and H. te Riele, *The Mertens conjecture revisited*, in Algorithmic Number Theory (ANTS VII), Lecture Notes in Comput. Sci. **4076**, Springer, Berlin, 2006.
[11] K. Matomäki and J. Teräväinen, *On the Möbius function in all short intervals*, J. Eur. Math. Soc. **25** (2023), no. 4, 1207–1225.
[12] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Stud. Adv. Math. **97**, Cambridge University Press, 2007.
[13] N. Ng, *The distribution of the summatory function of the Möbius function*, Proc. London Math. Soc. (3) **89** (2004), no. 2, 361–389.
[14] M. Nguyên Thê, *Area and inertial moment of Dyck paths*, Combin. Probab. Comput. **13** (2004), 697–716.
[15] A. M. Odlyzko and H. J. J. te Riele, *Disproof of the Mertens conjecture*, J. Reine Angew. Math. **357** (1985), 138–160.
[16] C. Richard, *On q-functional equations and excursion moments*, arXiv:math/0503198.
[17] L. Takács, *A Bernoulli excursion and its various applications*, Adv. Appl. Probab. **23** (1991), 557–585.
[18] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.
[19] A. Walfisz, *Weylsche Exponentialsummen in der neueren Zahlentheorie*, VEB Deutscher Verlag der Wissenschaften, Berlin, 1963.